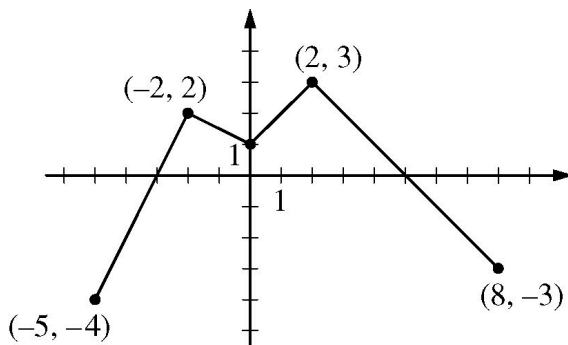


6-4

1.

Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) dt$.

- Find $g(0)$ and $g(-5)$.
- Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.
- On what intervals, if any, is the graph of g concave down? Give a reason for your answer.
- The function h is given by $h(x) = g(x^3 + 1)$. Find $h'(1)$. Show the work that leads to your answer.

Part A

The response can earn up to 2 points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) dt = 3$$

1 point is earned for $g(-5)$

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) dt = -10 + 3 = -7$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for $g(0)$

$$g(0) = 2 \cdot 0 + \int_{-2}^0 f(t) dt = 3$$

1 point is earned for $g(-5)$

6-4

$$g(-5) = 2 \cdot (-5) + \int_{-2}^{-5} f(t) dt = -10 + 3 = -7$$

Part B

The response can earn up to 3 points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$$g''(-2) = f'(-2) \text{ does not exist.}$$



0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$$g''(-2) = f'(-2) \text{ does not exist.}$$

Part C

The response can earn up to 1 point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

6-4



0	1
---	---

The response earns all of the following point:

1 point is earned for intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

Part D

The response can earn up to 3 points:

1 point is earned for chain rule

$$h'(x) = g'(x^3 + 1) \cdot 3x^2$$

1 point is earned for correct answer

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$



0	1	2	3
---	---	---	---

The response earns all of the following point:

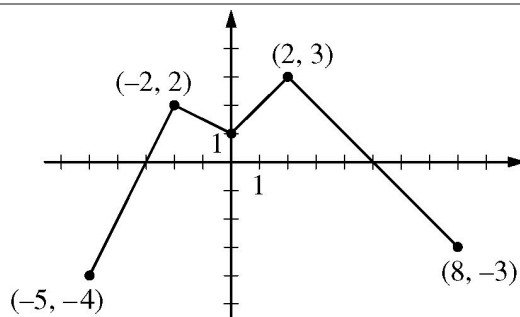
1 point is earned for chain rule

$$h'(x) = g'(x^3 + 1) \cdot 3x^2$$

1 point is earned for correct answer

$$\begin{aligned} h'(1) &= g'(2) \cdot 3 = (2 + f(2)) \cdot 3 \\ &= (2 + 3) \cdot 3 = 15 \end{aligned}$$

6-4

Graph of f

The continuous function f is defined on the interval $-5 \leq x \leq 8$. The graph of f , which consists of four line segments, is shown in the figure above. Let g be the function given by $g(x) = 2x + \int_{-2}^x f(t) dt$.

2. Find $g'(x)$ in terms of $f(x)$. For each of $g''(4)$ and $g''(-2)$, find the value or state that it does not exist.

Part B

The response can earn up to 3 points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

1 point: Stating $g''(-2)$ does not exist

$$g''(-2) = f'(-2) \text{ does not exist.}$$



0	1	2	3
---	---	---	---

The response earns all three of the following points:

1 point: $g'(x)$

$$g'(x) = 2 + f(x)$$

1 point: $g''(4)$

$$g''(x) = f'(x)$$

$$g''(4) = f'(4) = -1$$

6-4

1 point: Stating $g''(-2)$ does not exist

$g''(-2) = f'(-2)$ does not exist.

3. On what intervals, if any, is the graph of g concave down? Give a reason for your answer.

Part C

The response can earn up to 1 point:

1 point: For intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.



0	1
---	---

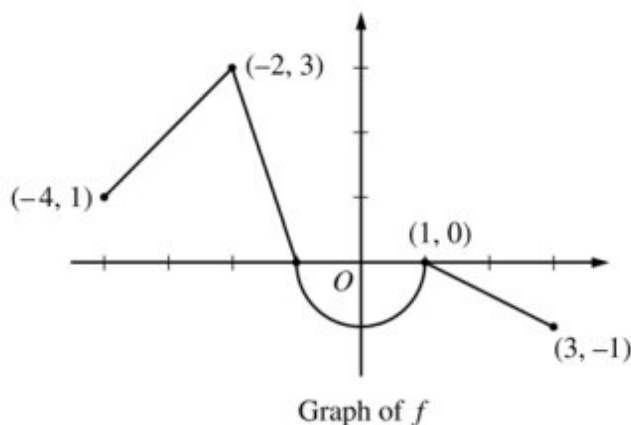
The response earn the following point:

1 point: For intervals and reason

The graph of g is concave down on the intervals $(-2, 0)$ and $(2, 8)$ since $g'(x) = 2 + f(x)$ decreases on those intervals.

6-4

4.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

- Find the values of $g(2)$ and $g(-2)$.
- For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2} \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(2)$

6-4

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \frac{\pi}{2} - \frac{3}{2} \end{aligned}$$

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$\begin{aligned} g'(x) &= f(x) \Rightarrow g'(-3) = f(-3) = 2 \\ g''(x) &= f'(x) \Rightarrow g''(-3) = f'(-3) = 1 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$\begin{aligned} g'(x) &= f(x) \Rightarrow g'(-3) = f(-3) = 2 \\ g''(x) &= f'(x) \Rightarrow g''(-3) = f'(-3) = 1 \end{aligned}$$

Part C

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

6-4

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.



0	1	2
---	---	---

The student response earns all of the following points:

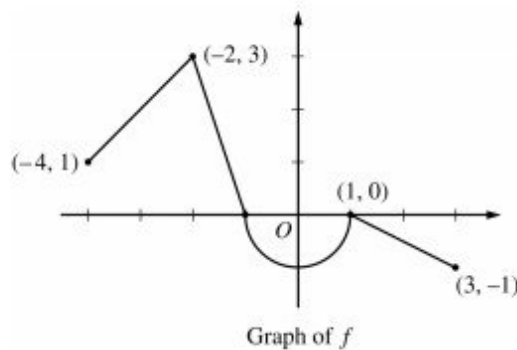
1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

6-4

5.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

Part B

One point is earned for $g'(-3)$

One point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



0	1	2
---	---	---

The student earns all of the following points:

One point is earned for $g'(-3)$

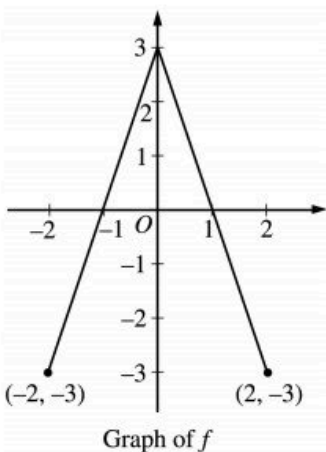
One point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

6-4

6.



The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

- Find $g(-1)$, $g'(-1)$, and $g''(-1)$.
- For what values of x in the open interval $(-2, 2)$ is g increasing? Explain your reasoning.
- For what values of x in the open interval $(-2, 2)$ is the graph of g concave down? Explain your reasoning.
- On the axes provided, sketch the graph of g on the closed interval $[-2, 2]$.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

6-4

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$

Part B

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the answer

g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

Part C

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

because $g'(x) = f(x)$ is decreasing on this interval.



0	1	2
---	---	---

The student response earns all of the following points:

6-4

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f(x) < 0$ on this interval.

or

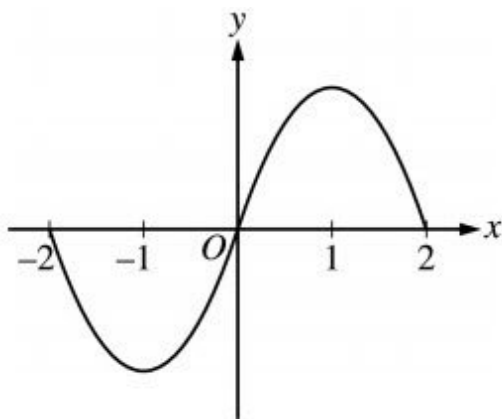
because $g'(x) = f(x)$ is decreasing on this interval.

Part D

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

<-1> vertical asymptote



0	1	2
---	---	---

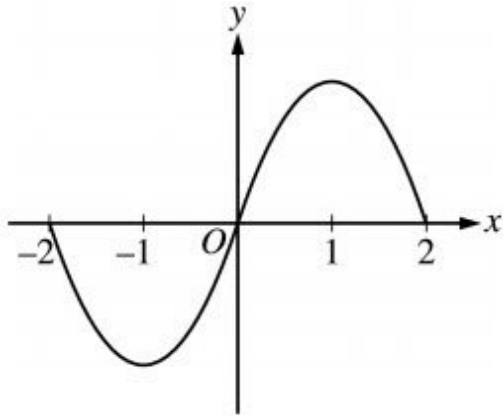
The student response earns all of the following points:

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

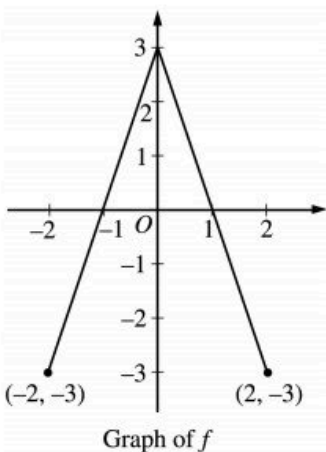
<-1> vertical asymptote

6-4



6-4

7.



The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

Find $g(-1)$, $g'(-1)$, and $g''(-1)$.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g''(-1)$

1 point is earned for $g'''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g''(-1)$

1 point is earned for $g'''(-1)$

6-4

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$

8.  Let $f(x) = \int_{-2}^{x^2-3x} e^{t^2} dt$. At what value of x is $f(x)$ a minimum?

(A) For no value of x

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$ ✓

(D) 2

(E) 3

9.  If F and f are differentiable functions such that $F(x) = \int_0^x f(t) dt$, and if $F(a) = -2$ and $F(b) = -2$ where $a < b$, which of the following must be true?

(A) $f(x)=0$ for some x such that $a < x < b$. ✓

(B) $f(x)>0$ for all x such that $a < x < b$.

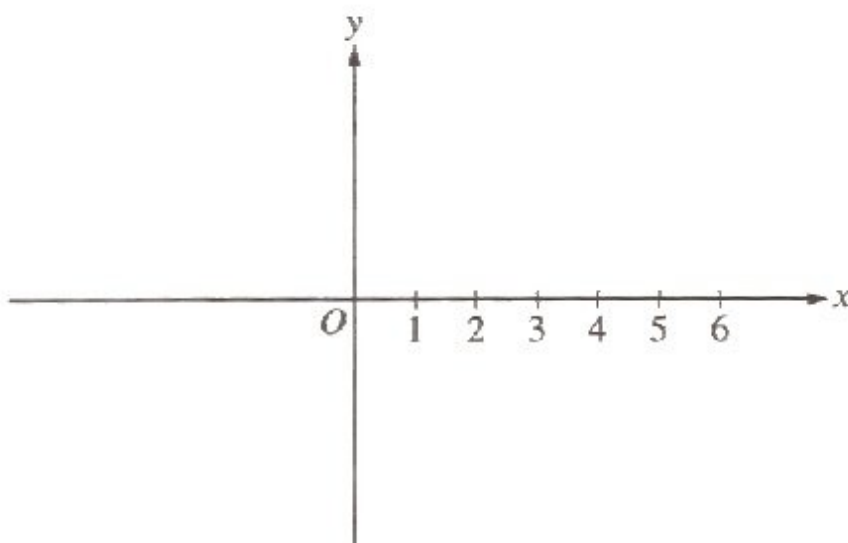
(C) $f(x)<0$ for all x such that $a < x < b$.

(D) $F(x) \leq 0$ for all x such that $a < x < b$.

6-4

10.  Let f be the function given by $f(x) = \sqrt{x - 3}$.

(a) On the axes provided below, sketch the graph of f and shade the region R enclosed by the graph of f , the x -axis, and the vertical line $x = 6$.



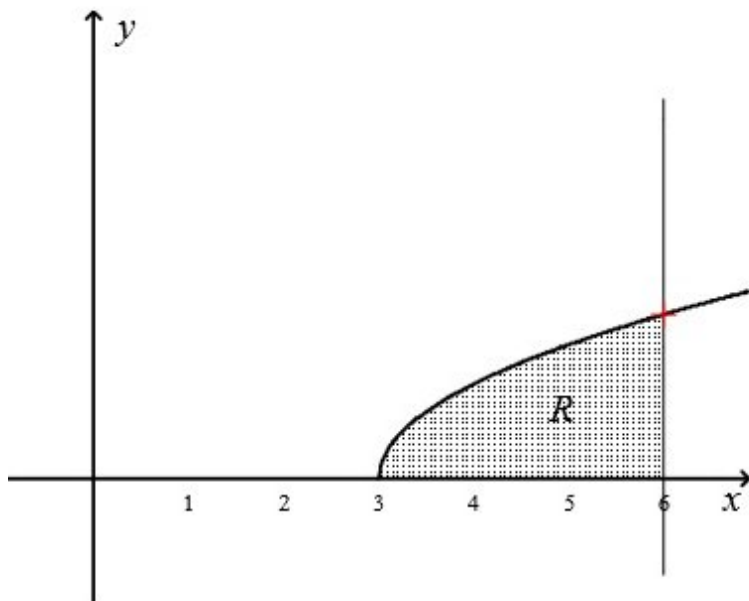
- (b) Find the area of the region R described in part (a).
- (c) Rather than using the line $x = 6$ as in part (a), consider the line $x = w$, where w can be any number greater than 3. Let $A(w)$ be the area of the region enclosed by the graph of f , the x -axis, and the vertical line $x = w$. Write an integral expression for $A(w)$.
- (d) Let $A(w)$ be as described in part (c). Find the rate of change of A with respect to w when $w = 6$.

Part A

1 point is earned for the correct graph of f , (domain is $x \geq 3$ goes through $(3,0)$, is increasing, positive, and concave down)

1 point is earned for the correct region relative to graph of f

6-4



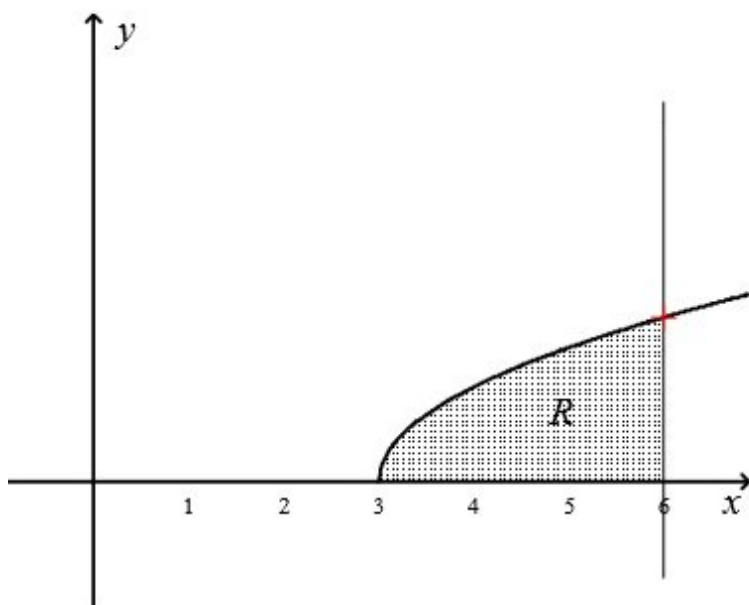
✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct graph of f , (domain is $x \geq 3$ goes through $(3, 0)$, is increasing, positive, and concave down)

1 point is earned for the correct region relative to graph of f



Part B

6-4

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} \, dx = \frac{2}{3}(x-3)^{3/2} \Big|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$\text{area} = \int_3^6 \sqrt{x-3} \, dx = \frac{2}{3}(x-3)^{3/2} \Big|_3^6$$

1 point is earned for the correct answer

0/1 if second point is not earned

$$= 2\sqrt{3} = 3.464$$

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} \, dx$$

6-4

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} \, dx$$

Part D

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

6-4

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$

Let f be the function given by $f(x) = \sqrt{x-3}$.

11. Rather than using the line $x = 6$ as in part (a), consider the line $x = w$, where w can be any number greater than 3. Let $A(w)$ be the area of the region enclosed by the graph of f , the x -axis, and the vertical line $x = w$. Write an integral expression for $A(w)$.

Part C

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} dx$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct limits

1 point is earned for the correct integrand

$$A(w) = \int_3^w \sqrt{x-3} dx$$

6-4

12. Let $A(w)$ be as described in part (c). Find the rate of change of A with respect to w when $w = 6$.
-

Part D

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct answer for $\frac{dA}{dw}$

$$\frac{dA}{dw} = \sqrt{w-3}$$

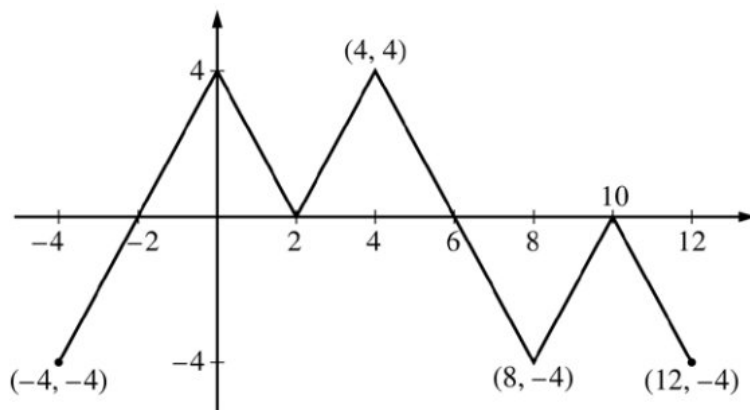
1 point is earned for the correct evaluation at 6

0/2 if $\frac{dA}{dw}$ is constant

$$\left. \frac{dA}{dw} \right|_{w=6} = \sqrt{3} = 1.732$$

6-4

13.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_2^x f(t) dt$.

- Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.
- Does the graph of g have a point of inflection at $x = 4$? Justify your answer.
- Find the absolute minimum value and the absolute maximum value of g on the interval $-4 \leq x \leq 12$. Justify your answers.
- For $-4 \leq x \leq 12$, find all intervals for which $g(x) \leq 0$.

Part A

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

Part B

6-4

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

Part C

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.



0	1	2	3	4	5
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The student response earns all of the following points:

6-4

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.

Part D

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.



0	1	2	3
---	---	---	---

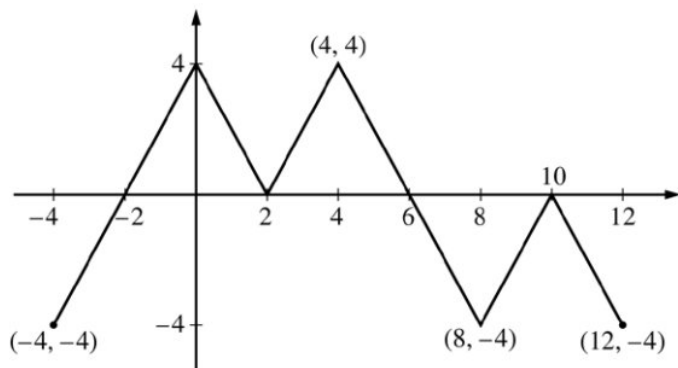
The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.

6-4

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_{-4}^x 2f(t) dt$.

14. Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.

Part A

1 point is earned for: $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for: answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

1 point is earned for: $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for: answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

6-4

15. Does the graph of g have a point of inflection at $x = 4$? Justify your answer.
-

Part B

1 point is earned for: answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0	1
---	---

Student response earns 1 of the following 1 point(s)

1 point is earned for: answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

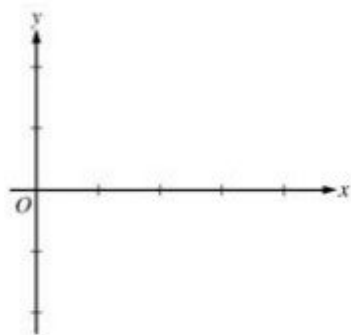
6-4

16.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

- (a) For $0 < x < 4$, find all values of x at which f has a relative extremum. Determine whether f has a relative maximum or a relative minimum at each of these values. Justify your answer.
- (b) On the axes provided, sketch the graph of a function that has all the characteristics of f .



- (c) Let g be the function defined by $g(x) = \int_1^x f(t) dt$ on the open interval $(0, 4)$. For $0 < x < 4$, find all values of x at which g has a relative extremum. Determine whether g has a relative maximum or a relative minimum at each of these values. Justify your answer.
- (d) For the function g defined in part (c), find all values of x , for $0 < x < 4$, at which the graph of g has a point of inflection. Justify your answer.

Part A

1 point is earned for the relative extremum at $x = 2$

1 point is earned for the relative maximum with justification

f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the relative extremum at $x = 2$

6-4

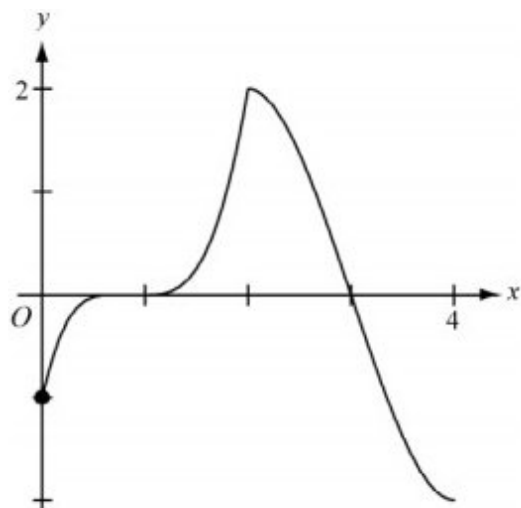
1 point is earned for the relative maximum with justification

f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.

Part B

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior



✓

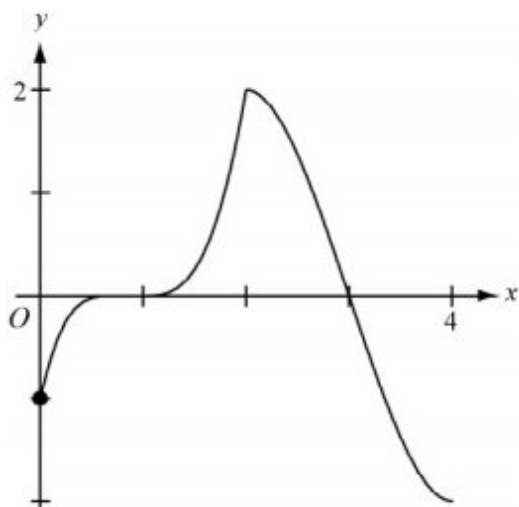
0	1	2
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The student response earns all of the following points:

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior

6-4

**Part C**

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.

Part D

1 point is earned for: $x = 2$

6-4

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.

6-4

17.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

Let g be the function defined by $g(x) = \int_1^x f(t) dt$ on the open interval $(0, 4)$. For $0 < x < 4$, find all values of x at which g has a relative extremum. Determine whether g has a relative maximum or a relative minimum at each of these values. Justify your answer.

Part C

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

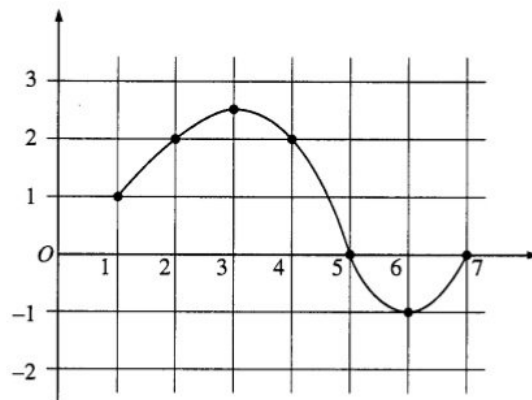
1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.

6-4

18.



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$

for $1 \leq x \leq 7$.

(a) Find $h(1)$.

(b) Find $h'(4)$.

(c) On what interval or intervals is the graph of h concave upward? Justify your answer.

(d) Find the value of x at which h has its minimum on the closed interval $[1, 7]$. Justify your answer.

Part A

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$



0

1

The student response earns all of the following points:

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

Part B

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$

6-4



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) =$$

Part C

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$$1 < x < 3 \text{ and } 6 < x < 7$$

1 point is earned for the correct justification

h is concave up when:

- h' is increasing, or
- f is increasing, or
- $h''(x) > 0$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct intervals where one point is earned for the first interval and one point is earned for the second interval

$$1 < x < 3 \text{ and } 6 < x < 7$$

1 point is earned for the correct justification

h is concave up when:

- h' is increasing, or
- f is increasing, or

6-4

- $h''(x) > 0$

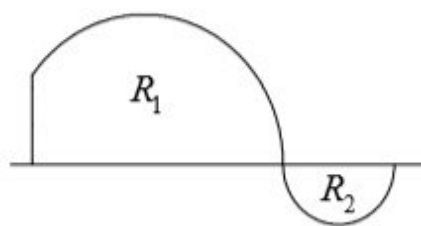
Part D

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$



0	1	2	3
---	---	---	---

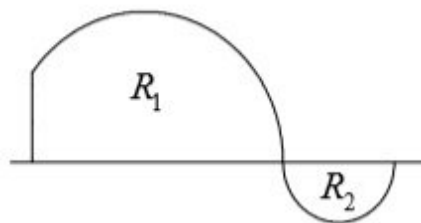
The student response earns all of the following points:

1 point is earned for the correct answer

1 point is earned for the correct justification where one point is earned for eliminating interior points and one point is earned for eliminating 7

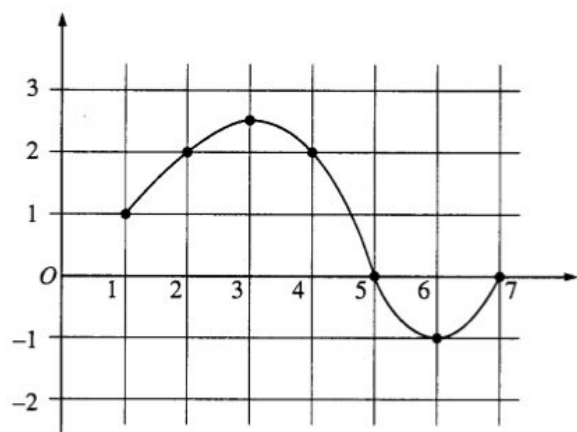
minimum at $x = 1$ because:

h increases on $[1, 5]$ and decreases on $[5, 7]$, so minimum is at an endpoint



6-4

$$h(7) = \text{area } R_1 - \text{area } R_2 > 0 \text{ and } h(1) = 0$$



The graph of a differentiable function f on the closed interval $[1, 7]$ is shown above. Let $h(x) = \int_1^x f(t) dt$ for $1 \leq x \leq 7$.

19. Find $h(1)$.

Part A

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$



0

1

The student response earns one of the following points:

1 point is earned for the correct answer

$$h(1) = \int_1^1 f(t) dt = 0$$

6-4

20. Find $h'(4)$.**Part B**1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) = 2$$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for correctly identifying $h'(x) = f(x)$

1 point is earned for the correct answer

$$h'(4) = f(4) = 2$$

21. Which of the following is an antiderivative of $f(x) = \sqrt{1+x^3}$?

(A) $\frac{2}{3}(1+x^3)^{\frac{3}{2}}$

(B) $\frac{\frac{2}{3}(1+x^3)^{\frac{3}{2}}}{3x^2}$

(C) $\int_0^{1+x^3} \sqrt{t} \, dt$

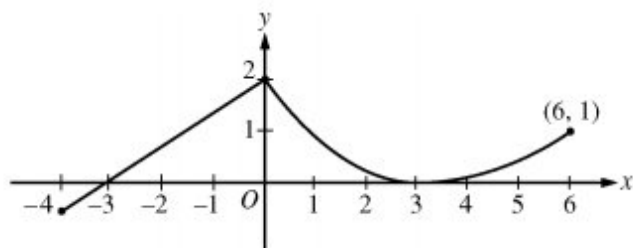
(D) $\int_0^{x^3} \sqrt{1+t} \, dt$

(E) $\int_0^x \sqrt{1+t^3} \, dt$



6-4

22.

Graph of f

A continuous function f is defined on the closed interval $-4 \leq x \leq 6$. The graph of f consists of a line segment and a curve that is tangent to the x -axis at $x=3$, as shown in the figure above. On the interval $0 < x < 6$, the function f is twice differentiable, with $f''(x) > 0$.

The function g is defined by $g(x) = \int_0^x f(t) dt$ for $-4 \leq x \leq 6$. On what intervals contained in $[-4, 6]$ is the graph of g concave up? Explain your reasoning.

Part D

1 point is earned for $g'(x)=f(x)$

1 point is earned for considering $g''(x)>0$

1 point is earned for the answer

$$g'(x)=f(x), g''(x)=f'(x)$$

$$g''(x)>0 \text{ when } f'(x)>0$$

This is true for $-4 < x < 0$ and $3 < x < 6$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x)=f(x)$

1 point is earned for considering $g''(x)>0$

1 point is earned for the answer

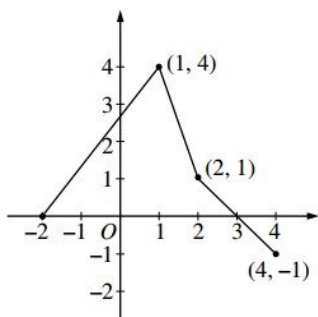
$$g'(x)=f(x), g''(x)=f'(x)$$

$$g''(x)>0 \text{ when } f'(x)>0$$

This is true for $-4 < x < 0$ and $3 < x < 6$.

6-4

23.



The graph of the function f , consisting of three line segments, is given above. Let $g(x) = \int_1^x f(t) dt$.

- Compute $g(4)$ and $g(-2)$.
- Find the instantaneous rate of change of g , with respect to x , at $x = 1$.
- Find the absolute minimum value of g on the closed interval $[-2, 4]$. Justify your answer.
- The second derivative of g is not defined at $x = 1$ and $x = 2$. How many of these values are x -coordinates of points of inflection of the graph of g ? Justify your answer.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

$$g(4) = \int_1^4 f(t) dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\frac{1}{2}(12) = -6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

6-4

$$g(4) = \int_1^4 f(t) dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\frac{1}{2}(12) = -6$$

Part B

1 point is earned for the answer

$$g'(1) = f(1) = 4$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$g'(1) = f(1) = 4$$

Part C

1 point is earned for interior analysis

1 point is earned for endpoint analysis

1 point is earned for the answer

g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for interior analysis

6-4

1 point is earned for endpoint analysis

1 point is earned for the answer

g is increasing on $[-2, 3]$ and decreasing on $[3, 4]$.

Therefore, g has absolute minimum at an endpoint of $[-2, 4]$.

Since $g(-2) = -6$ and $g(4) = \frac{5}{2}$, the absolute minimum value is -6 .

Part D

1 point is earned for the choice of $x = 1$ only

1 point is earned for showing $(1, g(1))$ is a point of inflection

1 point is earned for showing $(2, g(2))$ is not a point of inflection

One; $x = 1$

On $(-2, 1)$, $g''(x) = f'(x) > 0$

On $(1, 2)$, $g''(x) = f'(x) < 0$

On $(2, 4)$, $g''(x) = f'(x) < 0$

Therefore $(1, g(1))$ is a point of inflection and $(2, g(2))$ is not.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the choice of $x = 1$ only

1 point is earned for showing $(1, g(1))$ is a point of inflection

1 point is earned for showing $(2, g(2))$ is not a point of inflection

One; $x = 1$

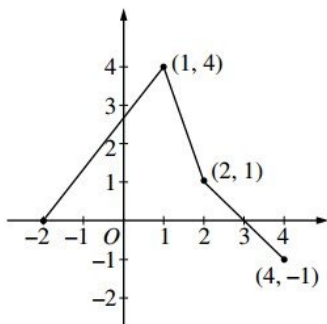
On $(-2, 1)$, $g''(x) = f'(x) > 0$

On $(1, 2)$, $g''(x) = f'(x) < 0$

On $(2, 4)$, $g''(x) = f'(x) < 0$

6-4

Therefore $(1, g(1))$ is a point of inflection and $(2, g(2))$ is not.



The graph of function f , consisting of three line segments, is given above. Let $g(x) = \int_1^x f(t) dt$.

24. Compute $g(4)$ and $g(-2)$.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

$$g(4) = \int_1^4 f(t) dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\frac{1}{2}(12) = -6$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g(-2)$

$$g(4) = \int_1^4 f(t) dt = \frac{3}{2} + 1 + \frac{1}{2} - \frac{1}{2} = \frac{5}{2}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\frac{1}{2}(12) = -6$$

6-4

25. Find the instantaneous rate of change of g , with respect to x , at $x = 1$.
-

Part B

1 point is earned for the answer

$$g'(1) = f(1) = 4$$



0	1
---	---

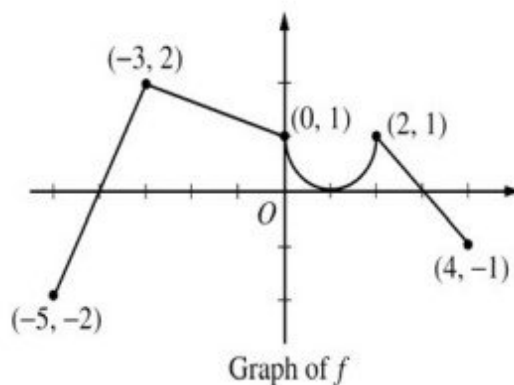
The student response earns all of the following points:

1 point is earned for the answer

$$g'(1) = f(1) = 4$$

6-4

26.



The graph of the function f shown above consists of a semicircle and three line segments. Let g be the function given by $g(x) = \int_{-3}^x f(t) dt$.

- Find $g(0)$ and $g'(0)$.
- Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Justify your answer.
- Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.
- Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

Part A

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2 + 1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(0)$

1 point is earned for $g'(0)$

6-4

$$g(0) = \int_{-3}^0 f(t) dt = \frac{1}{2}(3)(2+1) = \frac{9}{2}$$

$$g'(0) = f(0) = 1$$

Part B

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 3$

1 point is earned for justification

g has a relative maximum at $x = 3$.

This is the only x -value where $g' = f$ changes from positive to negative.

Part C

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13 - \pi}{2}$$

So the absolute minimum value of g is -1 .

6-4



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for identifying $x = -4$ as a candidate

1 point is earned for $g(-4) = -1$

1 point is earned for justification and answer

The only x -value where f changes from negative to positive is $x = -4$. The other candidates for the location of the absolute minimum value are the endpoints.

$$g(-5) = 0$$

$$g(-4) = \int_{-3}^{-4} f(t) dt = -1$$

$$g(4) = \frac{9}{2} + \left(2 - \frac{\pi}{2}\right) = \frac{13-\pi}{2}$$

So the absolute minimum value of g is -1 .

Part D

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$



0	1	2
---	---	---

The student response earns all of the following points:

Two points are earned for the correct values

<-1> each missing or extra value

$$x = -3, 1, 2$$

6-4

27.  The function f is given by $f(x) = \int_1^x \sqrt{t^3 + 2} dt$. What is the average rate of change of f over the interval $[0, 3]$?

- (A) 1.324
- (B) 1.497
- (C) 1.696
- (D) 2.266
- (E) 2.694



28. Let g be a function with first derivative given by $g'(x) = \int_0^x e^{-t^2} dt$. Which of the following must be true on the interval $0 < x < 2$?

- (A) g is increasing, and the graph of g is concave up.
- (B) g is increasing, and the graph of g is concave down.
- (C) g is decreasing, and the graph of g is concave up.
- (D) g is decreasing, and the graph of g is concave down.
- (E) g is decreasing, and the graph of g has a point of inflection on $0 < x < 2$.



6-4

29. Let $F(x) = \int_0^x \sin(t^2) dt$ for $0 \leq x \leq 3$.

On what intervals is F increasing?

Part B

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

1 point is earned for correctly analyzing sign of F' within $[0, 3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

1 point is earned for correctly using $F'(x) = \sin(x^2)$

$$F'(x) = \sin(x^2)$$

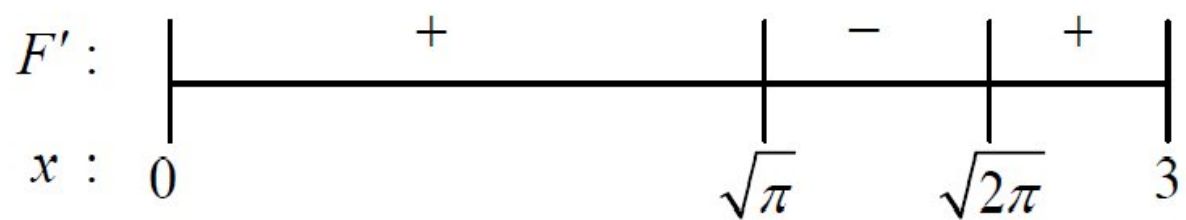
$$F'(x) = 0 \text{ when } x^2 = 0, \pi, 2\pi, \dots$$

1 point is earned for correctly finding zeros of F' (and eliminating extraneous zeros)

$$x = 0, \sqrt{\pi}, \sqrt{2\pi}$$

6-4

1 point is earned for correctly analyzing sign of F' within $[0,3]$



1 point is earned for correctly indicating F increases where student's F' is positive

F is increasing on $[0, \sqrt{\pi}]$ and on $[\sqrt{2\pi}, 3]$

6-4

30. The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

 Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.

Part C

1 point is earned for value $H'(17)$

2 points are earned for the meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for the meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

6-4

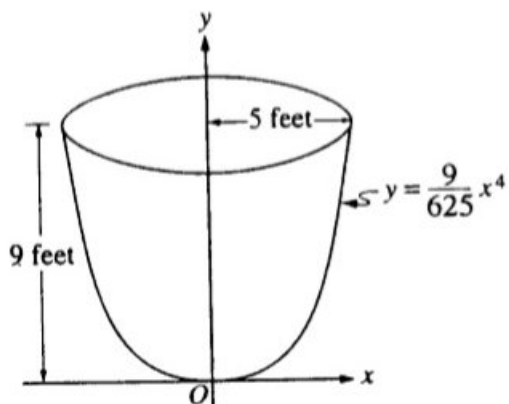
<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

6-4

31.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate all units of measure.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

6-4



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

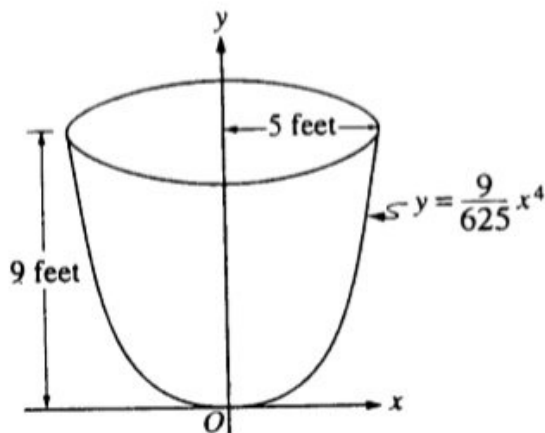
$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

6-4

32.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x=0$ to $x=5$ about the y -axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.

Let h be the depth, in feet, of oil in the tank. How fast is the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

Part C

1 point is earned for the correct volume as definite integral using h

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

6-4

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)



0	1	2	3	4	5
---	---	---	---	---	---

The student response earns five of the following points:

1 point is earned for the correct volume as definite integral using h

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct by $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

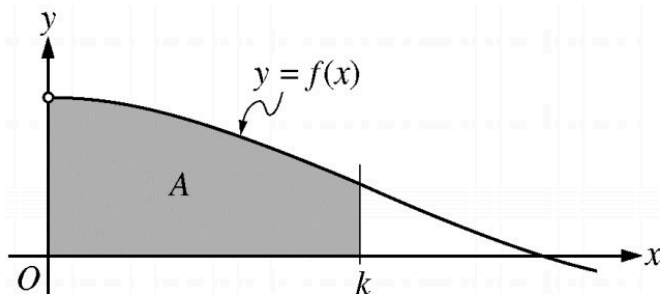
if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)

6-4

33.



The figure above shows the region A , which is bounded by the x - and y -axes, the graph of $f(x) = \frac{\sin x}{x}$ for $x > 0$, and the vertical line $x = k$. If k increases at a rate of $\pi/4$ units per second, how fast is the area of region A increasing when $k = \pi/6$?

(A) 0

(B) $3/4$ (C) $3/\pi$ (D) $\frac{\sqrt{3}}{2}$ (E) $2\sqrt{3}$ 

6-4

34. A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.

Write a polynomial expression for the position of the particle at any time $t \geq 0$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

< -1 > for error(s) in $t^3 - t^2 - t$

< -1 > for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

< -1 > for error(s) in $t^3 - t^2 - t$

< -1 > for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$

35. A particle moves along the x -axis with velocity given by $v(t) = 3t^2 - 4$ for time $t \geq 0$. If the particle is at position $x = -2$ at time $t = 0$, what is the position of the particle at the time $t = 3$?

(A) 13

(B) 15

(C) 16

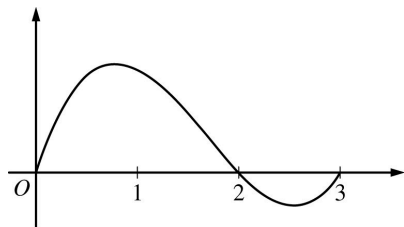
(D) 17

(E) 25



6-4

36.

Graph of f

The graph of the differentiable function f is shown in the figure above. Let h be the function defined by $h(x) = \int_0^x f(t) dt$. Which of the following correctly orders $h(2)$, $h'(2)$, and $h''(2)$?

(A) $h(2) < h'(2) < h''(2)$

(B) $h'(2) < h(2) < h''(2)$

(C) $h'(2) < h''(2) < h(2)$

(D) $h''(2) < h(2) < h'(2)$

(E) $h''(2) < h'(2) < h(2)$



6-4

37. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.

(a) Find $f'(x)$ and $g'(x)$.

(b) Write an equation for the line tangent to the graph of $y = g(x)$ at $x = \pi$.

(c) Write, but do not evaluate, an integral expression that represents the maximum value of g on the interval $0 \leq x \leq \pi$. Justify your answer.

Part A

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$f'(x) = 3\sqrt{4 + (3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4 + (3\sin x)^2} \cdot \cos x$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$f'(x) = 3\sqrt{4 + (3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4 + (3\sin x)^2} \cdot \cos x$$

Part B

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

6-4



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(\pi)$ or $g'(\pi)$

1 point is earned for the tangent line equation

$$g(\pi) = 0, g'(\pi) = -6$$

$$\text{Tangent line: } y = -6(x - \pi)$$

Part C

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4 + t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4 + t^2} dt$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $g'(x) = 0$

1 point is earned if justifies maximum at $\frac{\pi}{2}$

1 point is earned for integral expression at $g\left(\frac{\pi}{2}\right)$

6-4

For $0 < x < \pi$, $g'(x) = 0$ only at $x = \frac{\pi}{2}$.

$$g(0) = g(\pi) = 0$$

$$g\left(\frac{\pi}{2}\right) = \int_0^3 \sqrt{4+t^2} dt > 0$$

The maximum value of g on $[0, \pi]$ is $\int_0^3 \sqrt{4+t^2} dt$.

38. The functions f and g are given by $f(x) = \int_0^{3x} \sqrt{4+t^2} dt$ and $g(x) = f(\sin x)$.

Find $f'(x)$ and $g'(x)$.

Part A

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

$$f'(x) = 3\sqrt{4+(3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4+(3\sin x)^2} \cdot \cos x$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for $f'(x)$

2 points are earned for $g'(x)$

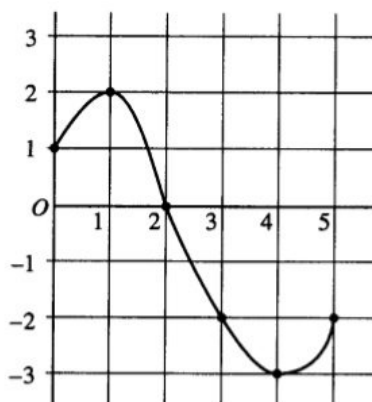
$$f'(x) = 3\sqrt{4+(3x)^2}$$

$$g'(x) = f'(\sin x) \cdot \cos x$$

$$= 3\sqrt{4+(3\sin x)^2} \cdot \cos x$$

6-4

39.

Graph of f

Let f be a function whose domain is the closed interval $[0, 5]$. The graph of f is shown above.

Let $h(x) = \int_0^{\frac{x}{2}+3} f(t) dt$.

Find $h'(2)$.

Part B

2 points are earned for the correct answer for $h'(x) = f(\frac{x}{2} + 3) \cdot \frac{1}{2}$

< -1 > no chain rule

1 point is earned for the correct answer, using $f(4)$

$$h'(x) = f(\frac{x}{2} + 3) \cdot \frac{1}{2}$$

$$h'(2) = f(4) \cdot \frac{1}{2} = -\frac{3}{2}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct answer for $h'(x) = f(\frac{x}{2} + 3) \cdot \frac{1}{2}$

< -1 > no chain rule

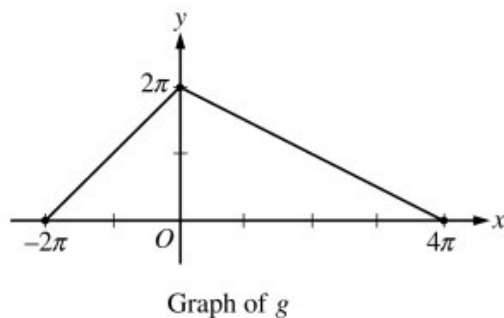
1 point is earned for the correct answer, using $f(4)$

$$h'(x) = f(\frac{x}{2} + 3) \cdot \frac{1}{2}$$

$$h'(2) = f(4) \cdot \frac{1}{2} = -\frac{3}{2}$$

6-4

40.



Let g be the piecewise-linear function defined on $[-2\pi, 4\pi]$ whose graph is given above, and let $f(x) = g(x) - \cos\left(\frac{x}{2}\right)$.

- (a) Find $\int_{-2\pi}^{4\pi} f(x) dx$. Show the computations that lead to your answer.
- (b) Find all x -values in the open interval $(-2\pi, 4\pi)$ for which f has a critical point.
- (c) Let $h(x) = \int_0^{3x} g(t) dt$. Find $h'\left(-\frac{\pi}{3}\right)$.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_{-2\pi}^{4\pi} f(x) dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right) \right) dx \\ &= 6\pi^2 - \left[2 \sin\left(\frac{x}{2}\right) \right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

6-4

$$\begin{aligned}
 \int_{-2\pi}^{4\pi} f(x) dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right) \right) dx \\
 &= 6\pi^2 - \left[2 \sin\left(\frac{x}{2}\right) \right]_{x=-2\pi}^{x=4\pi} \\
 &= 6\pi^2
 \end{aligned}$$

Part B

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

6-4

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$

6-4

41. For $x > 0$, $\frac{d}{dx} \left(\int_0^{2x} \ln(t^3 + 1) dt \right) =$

(A) $\ln(x^3 + 1)$

(B) $\ln(8x^3 + 1)$

(C) $2 \ln(x^3 + 1)$

(D) $2 \ln(8x^3 + 1)$ ✓

(E) $24x^2 \ln(8x^3 + 1)$

42. For $x > 0$, $\frac{d}{dx} \int_1^{\sqrt{x}} \frac{1}{1+t^2} dt =$

(A) $\frac{1}{2\sqrt{x}(1+x)}$ ✓

(B) $\frac{1}{2\sqrt{x}(1+\sqrt{x})}$

(C) $\frac{1}{1+x}$

(D) $\frac{\sqrt{x}}{1+x}$

(E) $\frac{1}{1+\sqrt{x}}$

43. For $x > 0$, $\frac{d}{dx} \int_1^{\sqrt{x}} \frac{1}{1+t^2} dt =$

(A) $\frac{1}{2\sqrt{x}(1+x)}$ ✓

(B) $\frac{1}{2\sqrt{x}(1+\sqrt{x})}$

(C) $\frac{1}{1+x}$

(D) $\frac{\sqrt{x}}{1+x}$

(E) $\frac{1}{1+\sqrt{x}}$

6-4

44.  Let $f(x) = \int_0^{x^2} \sin t \, dt$. At how many points in the closed interval $[0, \sqrt{\pi}]$ does the instantaneous rate of change of f equal the average rate of change of f on that interval?

(A) Zero

(B) One

(C) Two



(D) Three

(E) Four

45. If f is the function given by $f(x) = \int_4^{2x} \sqrt{t^2 - t} \, dt$, then $f'(2) =$

(A) 0

(B) $\frac{7}{2\sqrt{12}}$ (C) $\sqrt{2}$ (D) $\sqrt{12}$ (E) $2\sqrt{12}$ 

6-4

46.  The rate at which people enter an amusement park on a given day is modeled by the function E defined by

$$E(t) = \frac{15600}{(t^2 - 24t + 160)}.$$

The rate at which people leave the same amusement park on the same day is modeled by the function L defined by

$$L(t) = \frac{9890}{(t^2 - 38t + 370)}.$$

Both $E(t)$ and $L(t)$ are measured in people per hour and time t is measured in hours after midnight. These functions are valid for $9 \leq t \leq 23$, the hours during which the park is open. At time $t = 9$, there are no people in the park.

(a) How many people have entered the park by 5:00 P.M. ($t = 17$)? Round your answer to the nearest whole number.

(b) The price of admission to the park is \$15 until 5:00 P.M. ($t = 17$). After 5:00 P.M., the price of admission to the park is \$11. How many dollars are collected from admissions to the park on the given day? Round your answer to the nearest whole number.

(c) Let $H(t) = \int_9^t (E(x) - L(x))dx$ for $9 \leq t \leq 23$. The value of $H(17)$ to the nearest whole number is 3725. Find the value of $H'(17)$, and explain the meaning of $H(17)$ and $H'(17)$ in the context of the amusement park.

(d) At what time t , for $9 \leq t \leq 23$, does the model predict that the number of people in the park is a maximum?

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t)dt = 6004.270$$

6004 people entered the park by 5 pm.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

6-4

1 point is earned for integrand

1 point is earned for the answer

$$\int_9^{17} E(t) dt = 6004.270$$

6004 people entered the park by 5 pm.

Part B

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the setup

$$15 \int_9^{17} E(t) dt + 11 \int_{17}^{23} E(t) dt = 104048.165$$

The amount collected was \$104,048.

or

6-4

$$\int_{17}^{23} E(t) dt = 1271.283$$

1271 people entered the park between 5 pm and 11 pm, so the amount collected was

$$\$15 \cdot (6004) + \$11 \cdot (1271) = \$104,041.$$

Part C

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for value $H'(17)$

2 points are earned for meanings

1: meaning of $H(17)$

1: meaning of $H'(17)$

<-1> if no reference to $t = 17$

$$H'(17) = E(17) - L(17) = -380.281$$

There were 3725 people in the park at $t = 17$. The number of people in the park was decreasing at the rate of approximately 380 people/hr at time $t = 17$.

6-4**Part D**

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $E(t) - L(t) = 0$

1 point is earned for the answer

$$H'(t) = E(t) - L(t) = 0$$

$$t = 15.794 \text{ or } 15.795$$

6-4

47. For all values of x , the continuous function f is positive and decreasing. Let g be the function given by $g(x) = \int_2^x f(t) dt$. Which of the following could be a table of values for g ?

(A)

x	$g(x)$
1	-2
2	0
3	1



(B)

x	$g(x)$
1	-2
2	0
3	3

(C)

x	$g(x)$
1	1
2	0
3	-2

(D)

x	$g(x)$
1	2
2	0
3	-1

(E)

x	$g(x)$
1	3
2	0
3	2

6-4

48. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.

At what time t , for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \quad \text{when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

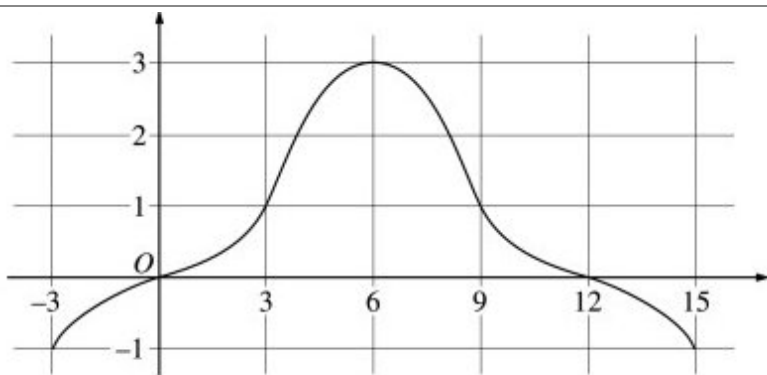
1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \quad \text{when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.

6-4

Graph of f

The graph of a differentiable function f on the closed interval $[-3, 15]$ is shown in the figure above. The graph of f has a horizontal tangent line at $x = 6$. Let $g(x) = 5 + \int_6^x f(t)dt$ for $-3 \leq x \leq 15$.

49. On what intervals is the graph of g concave down? Justify your answer.

Part C

1 point is earned for the interval

1 point is earned for the justification

The graph of g is concave down on $(6, 15)$ since $g' = f$ is decreasing on this interval



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the justification

The graph of g is concave down on $(6, 15)$ since $g' = f$ is decreasing on this interval

6-4

50. On what intervals is g decreasing? Justify your answer.

Part B

1 point is earned for: $[-3, 0]$

1 point is earned for: $[12, 15]$

1 point is earned for the justification

g is decreasing on $[-3, 0]$ and $[12, 15]$ since $g'(x) = f(x) < 0$ for $x < 0$ and $x > 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $[-3, 0]$

1 point is earned for: $[12, 15]$

1 point is earned for the justification

g is decreasing on $[-3, 0]$ and $[12, 15]$ since $g'(x) = f(x) < 0$ for $x < 0$ and $x > 12$.

6-4

51. Find $g(6)$, $g'(6)$, and $g''(6)$.**Part A**1 point is earned for: $g(6)$ 1 point is earned for: $g'(6)$ 1 point is earned for: $g''(6)$

$$g(6) = 5 + \int_6^6 f(t)dt = 5$$

$$g'(6) = f(6) = 3$$

$$g''(6) = f''(6) = 0$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

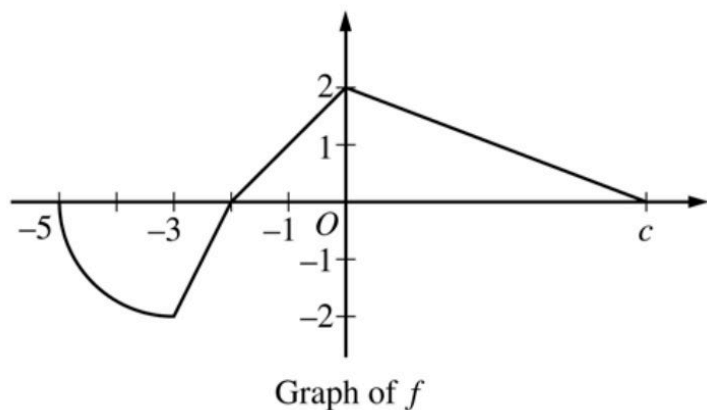
1 point is earned for: $g(6)$ 1 point is earned for: $g'(6)$ 1 point is earned for: $g''(6)$

$$g(6) = 5 + \int_6^6 f(t)dt = 5$$

$$g'(6) = f(6) = 3$$

$$g''(6) = f''(6) = 0$$

6-4



The function f is defined on the interval $-5 \leq x \leq c$, where $c > 0$ and $f(c) = 0$. The graph of f , which consists of three line segments and a quarter of a circle with center $(-3, 0)$ and radius 2, is shown in the figure above.

52. For $-5 \leq x \leq c$, let g be the function defined by $g(x) = \int_{-1}^x f(t) dt$. Find the x -coordinate of each point of inflection of the graph of g . Justify your answer.

Part B

1 point(s) earned for: $g'(x) = f(x)$

$$g'(x) = f(x)$$

1 point(s) earned for: identifies $x = -3$ and $x = 0$

The graph of g has a point of inflection at $x = -3$ because $g' = f$ changes from decreasing to increasing at this point

1 point(s) earned for: justification

The graph of g has a point of inflection at $x = 0$ because $g' = f$ changes from increasing to decreasing at this point



0	1	2	3
---	---	---	---

Student response earns 3 of the following 3 point(s)

1 point(s) earned for: $g'(x) = f(x)$

$$g'(x) = f(x)$$

1 point(s) earned for: identifies $x = -3$ and $x = 0$

6-4

The graph of g has a point of inflection at $x = -3$ because $g' = f$ changes from decreasing to increasing at this point

1 point(s) earned for: justification

The graph of g has a point of inflection at $x = 0$ because $g' = f$ changes from increasing to decreasing at this point

6-4

53. For $-5 \leq x \leq c$, let g be the function defined by $g(x) = \int_{-1}^x f(t)dt$. Find the x -coordinate of each point of inflection of the graph of g . Justify your answer.

Part B

3 points can be earned;

1 point for $g'(x) = f(x)$

1 point for $x = -3$ and $x = 0$

1 point for justification

$$g'(x) = f(x)$$

The graph of g has a point of inflection at $x = -3$ because

$g' = f$ changes from decreasing to increasing at this point.

The graph of g has a point of inflection at $x = 0$ because

$g' = f$ changes from increasing to decreasing at this point.



0	1	2	3
---	---	---	---

The student response earns none of the following points:

3 points can be earned;

1 point for $g'(x) = f(x)$

1 point for $x = -3$ and $x = 0$

1 point for justification

$$g'(x) = f(x)$$

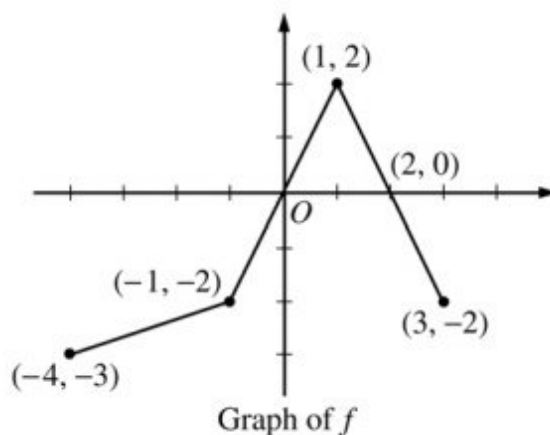
The graph of g has a point of inflection at $x = -3$ because

6-4

$g' = f$ changes from decreasing to increasing at this point.

The graph of g has a point of inflection at $x = 0$ because

$g' = f$ changes from increasing to decreasing at this point.



The graph of the function f above consists of three line segments.

54. For the function h defined in part (c), find all intervals on which h is decreasing. Explain your reasoning.

Part D

1 point is earned for interval

1 point is earned for reason

h is decreasing on $[0, 2]$

$h' = -f < 0$ when $f > 0$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interval

1 point is earned for reason

h is decreasing on $[0, 2]$

$h' = -f < 0$ when $f > 0$

6-4

55. Let g be the function given by $g(x) = \int_{-4}^x f(t) dt$. For each of $g(-1)$, $g'(-1)$, and $g''(-1)$, find the value or state that it does not exist.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t) dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t) dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.

6-4

56. Let h be the function given by $h(x) = \int_x^3 f(t)dt$. Find all values of x in the closed interval $-4 \leq x \leq 3$ for which $h(x)=0$.
-

Part C

2 point is earned for the correct values

(-1) each missing or extra value

$x=-1,1,3$



0	1	2
---	---	---

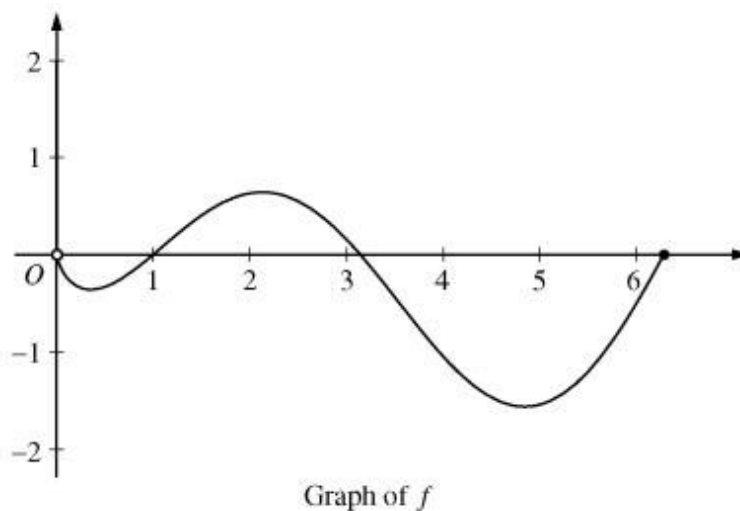
The student response earns all of the following points:

2 point is earned for the correct values

(-1) each missing or extra value

$x=-1,1,3$

6-4



Let f be the function given by $f(x) = (\ln x)(\sin x)$. The figure above shows the graph of f for $0 < x \leq 2\pi$. The function g is defined by $g(x) = \int_1^x f(t) dt$ for $0 < x \leq 2\pi$.

57. On what intervals, if any, is g increasing? Justify your answer.

Part B

1 point is earned for the interval

1 point is earned for the reason

Since $g'(x) = f(x)$, g is increasing on the interval $1 \leq x \leq \pi$ because $f(x) > 0$ for $1 < x < \pi$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the reason

Since $g'(x) = f(x)$, g is increasing on the interval $1 \leq x \leq \pi$ because $f(x) > 0$ for $1 < x < \pi$.

6-4

58. Find $g(1)$ and $g'(1)$.**Part A**1 point is earned for $g(1)$ 1 point is earned for $g'(1)$

$$g(1) = \int_1^1 f(t)dt = 0 \text{ and } g'(1) = f(1) = 0$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(1)$ 1 point is earned for $g'(1)$

$$g(1) = \int_1^1 f(t)dt = 0 \text{ and } g'(1) = f(1) = 0$$

59. If $F(x) = \int_0^x \sqrt{t^3 + 1}dt$, then $F'(2) =$

(A) -3

(B) -2

(C) 2

(D) 3

(E) 18



6-4

60.  Let g be the function given by $g(x) = \int_0^x \sin(t^2) dt$ for $-1 \leq x \leq 3$. On which of the following intervals is g decreasing?

- (A) $-1 \leq x \leq 0$
(B) $0 \leq x \leq 1.772$
(C) $1.253 \leq x \leq 2.171$
(D) $1.772 \leq x \leq 2.507$ ✓
(E) $2.802 \leq x \leq 3$

61. For $x > 0$, $\int \left(\frac{1}{x} \int_1^x \frac{du}{u} \right) dx =$

- (A) $\frac{1}{x^3} + C$
(B) $\frac{8}{x^4} - \frac{2}{x^2} + C$
(C) $\ln(\ln x) + C$
(D) $\frac{\ln(x^2)}{2} + C$
(E) $\frac{(\ln x)^2}{2} + C$ ✓

62. $\frac{d}{dx} \int_2^x \sqrt{1+t^2} dt =$

- (A) $\frac{x}{\sqrt{1+x^2}}$
(B) $\sqrt{1+x^2} - 5$
(C) $\sqrt{1+x^2}$ ✓
(D) $\frac{x}{\sqrt{1+x^2}} - \frac{1}{\sqrt{5}}$
(E) $\frac{1}{2\sqrt{1+x^2}} - \frac{1}{2\sqrt{5}}$

63. $\frac{d}{dx} \left(\int_0^{x^2} \sin(t^3) dt \right) =$

- (A) $-\cos(x^6)$
(B) $\sin(x^3)$
(C) $\sin(x^6)$
(D) $2x \sin(x^3)$
(E) $2x \sin(x^6)$ ✓

6-4

64. For all $x > 1$, if $f(x) = \int_1^x \frac{1}{t} dt$, then $f'(x) =$
- (A) 1
- (B) $\frac{1}{x}$ ✓
- (C) $\ln x - 1$
- (D) $\ln x$
- (E) e^x
65. Let h be the function defined by $h(x) = \int_{\pi/4}^x \sin^2 t \, dt$. Which of the following is an equation for the line tangent to the graph of h at the point where $x = \frac{\pi}{4}$?
- (A) $y = 1/2$
- (B) $y = \sqrt{2}x$
- (C) $y = x - \frac{\pi}{4}$
- (D) $y = \frac{1}{2}\left(x - \frac{\pi}{4}\right)$ ✓
- (E) $y = \left(\frac{\sqrt{2}}{2}\right)\left(x - \frac{\pi}{4}\right)$
66. $\frac{d}{dx} \left(\int_0^{x^3} \ln(t^2 + 1) dt \right) =$
- (A) $\frac{2x^3}{x^6 + 1}$
- (B) $\frac{3x^2}{x^6 + 1}$
- (C) $\ln(x^6 + 1)$
- (D) $2x^3 \ln(x^6 + 1)$
- (E) $3x^2 \ln(x^6 + 1)$ ✓
67.  Water is pumped out of a lake at the rate $R(t) = 12\sqrt{\frac{t}{t+1}}$ cubic meters per minute, where t is measured in minutes. How much water is pumped from time $t = 0$ to $t = 5$?
- (A) 9.439 cubic meters
- (B) 10.954 cubic meters
- (C) 43.816 cubic meters
- (D) 47.193 cubic meters ✓
- (E) 54.772 cubic meters

6-4

68. Which of the following is an equation for the line tangent to the graph of $y = 3 - \int_{-1}^x e^{-t^3} dt$ at the point where $x = -1$?

(A) $y - 3 = -3e(x + 1)$

(B) $y - 3 = -e(x + 1)$ ✓

(C) $y - 3 = 0$

(D) $y - 3 = -1/e(x + 1)$

(E) $y - 3 = 3e(x + 1)$

69.  Let f be the function given by $f(x) = \int_{1/3}^x \cos\left(\frac{1}{t^2}\right) dt$ for $\frac{1}{3} \leq x \leq 1$. At which of the following values of x does f attain a relative maximum?

(A) 0.357 and 0.798

(B) 0.4 and 0.564

(C) 0.4 only

(D) 0.461 ✓

(E) 0.999

70. If $F(x) = \int_4^{x^2} \sqrt{t} dt$ for all real numbers $x > 0$, then $F'(x) =$

(A) $-\frac{1}{2x}$

(B) $\sqrt{x}$

(C) x

(D) $2x^2$ ✓

(E) $\frac{2x^3 - 16}{3}$

71. Let g be the function defined by $g(x) = \int_{-1}^x \frac{t^3 - t^2 - 6t}{\sqrt{t^2 + 7}} dt$. On which of the following intervals is g decreasing?

(A) $x \leq -2$ and $0 \leq x \leq 3$ ✓

(B) $x \leq -2$ and $x \geq 3$

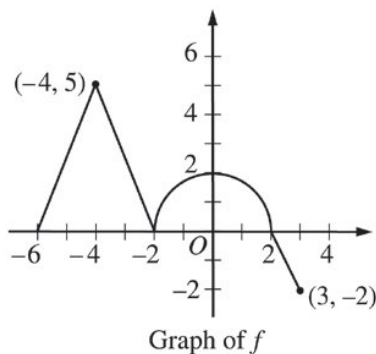
(C) $-2 \leq x \leq 0$ and $x \geq 3$

(D) $-2 \leq x \leq 3$

(E) $x \leq -1$

6-4

72.



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

Find $g'(0)$.

Part B

The response can earn up to 1 point:

1 point: For correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$



0

1

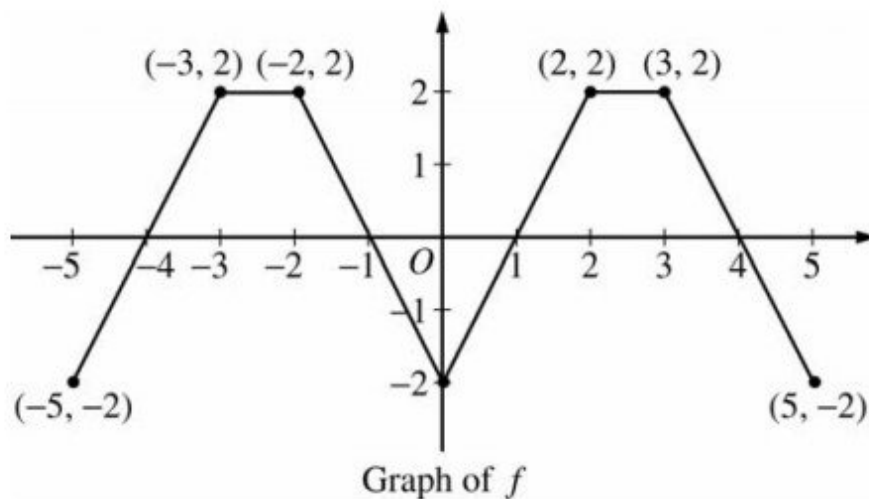
The response earns the following point:

1 point: For correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$

6-4

73.



The graph of the function f shown above consists of six line segments. Let g be the function given by

$$g(x) = \int_0^x f(t) dt.$$

Find $g(4)$, $g'(4)$, and $g''(4)$.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

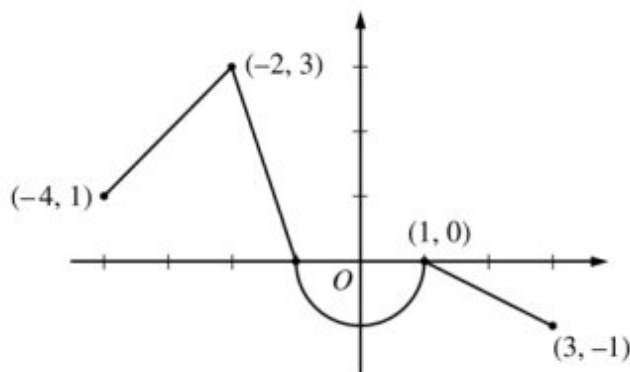
6-4

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$

74.

Graph of f

Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.

Part B

1 point is earned for $g'(-3)$

1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$

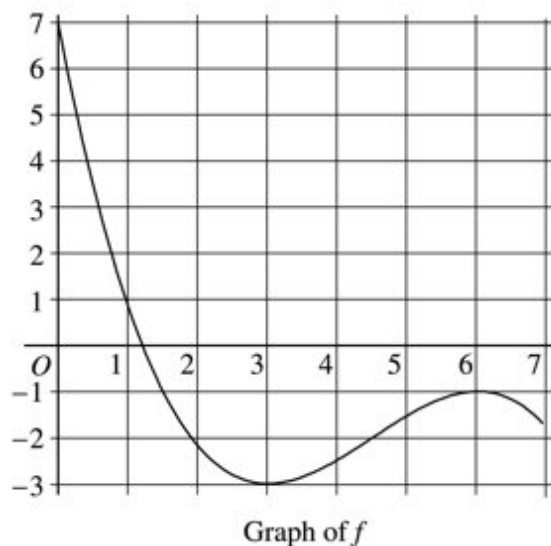
1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

6-4

75.



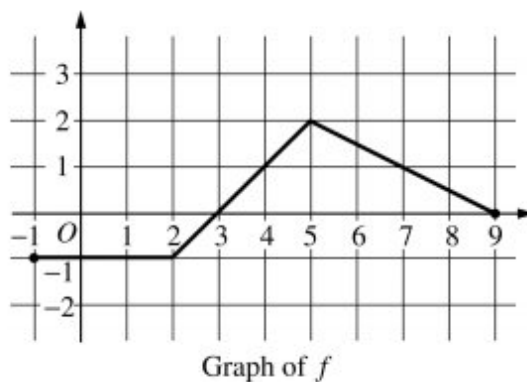
The graph of the function f shown in the figure above has horizontal tangents at $x = 3$ and $x = 6$. If

$$g(x) = \int_0^{2x} f(t) dt \text{ what is the value of } g'(3)?$$

- (A) 0
- (B) -1
- (C) -2
- (D) -3
- (E) -6



76.



The graph of the piecewise linear function f is shown above. Let h be the function given by

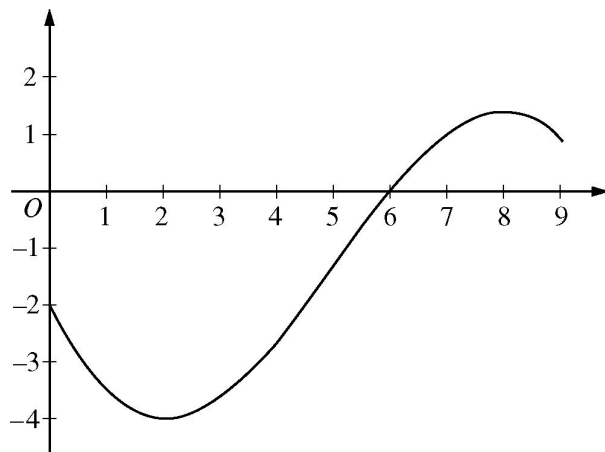
$$h(x) = \int_{-1}^x f(t) dt. \text{ On which of the following intervals is } h \text{ increasing?}$$

- (A) $[-1, 3]$
- (B) $[0, 5]$
- (C) $[2, 5]$ only
- (D) $[2, 9]$
- (E) $[3, 9]$ only



6-4

77.

Graph of f

The graph of a differentiable function f is shown above. If $h(x) = \int_0^x f(t)dt$, which of the following is true?

(A) $h(6) < h'(6) < h''(6)$



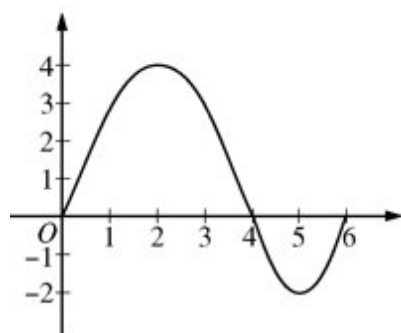
(B) $h(6) < h''(6) < h'(6)$

(C) $h'(6) < h(6) < h''(6)$

(D) $h''(6) < h(6) < h'(6)$

(E) $h''(6) < h'(6) < h(6)$

78.

Graph of f

The graph of the function f shown above has horizontal tangents at $x = 2$ and $x = 5$. Let g be the function defined by $g(x) = \int_0^x f(t)dt$. For what values of x does the graph of g have a point of inflection?

(A) 2 only

(B) 4 only

(C) 2 and 5 only



(D) 2, 4, and 5

(E) 0, 4, and 6

6-4

79. On the graph of $y = f(x)$, the slope at any point (x, y) is twice the value of x . If $f(2) = 3$, what is the value of $f(3)$?

(A) 6

(B) 7

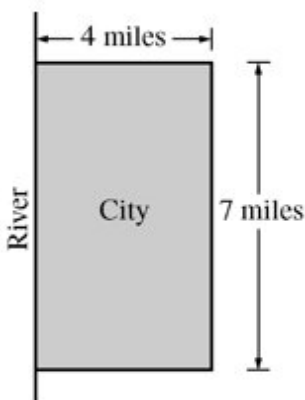
(C) 8

(D) 9

(E) 10



80.



A city located beside a river has a rectangular boundary as shown in the figure above. The population density of the city at any point along a strip x miles from the river's edge is $f(x)$ persons per square mile. Which of the following expressions gives the population of the city?

(A) $\int_0^4 f(x) dx$

(B) $7 \int_0^4 f(x) dx$

(C) $28 \int_0^4 f(x) dx$

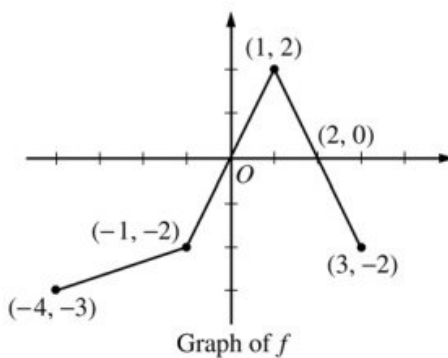
(D) $\int_0^7 f(x) dx$

(E) $4 \int_0^7 f(x) dx$



6-4

81.



The graph of the function f above consists of three line segments.

- (a) Let g be the function given by $g(x) = \int_{-4}^x f(t) dt$. For each of $g(-1)$, $g'(-1)$, and $g''(-1)$, find the value or state that it does not exist.
- (b) For the function g defined in part (a), find the x -coordinate of each point of inflection of the graph of g on the open interval $-4 < x < 3$. Explain your reasoning.
- (c) Let h be the function given by $h(x) = \int_x^3 f(t) dt$. Find all values of x in the closed interval $-4 \leq x \leq 3$ for which $h(x) = 0$.
- (d) For the function h defined in part (c), find all intervals on which h is decreasing. Explain your reasoning.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t) dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

6-4

1 point is earned for $g''(-1)$

$$g(-1) = \int_{-4}^{-1} f(t) dt = -\frac{1}{2}(3)(5) = -\frac{15}{2}$$

$$g'(-1) = f(-1) = -2$$

$g''(-1)$ does not exist because f is not differentiable at $x = -1$.

Part B

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$$x = 1$$

$g' = f$ changes from increasing to decreasing at $x = 1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $x = 1$ (only)

1 point is earned for the reason

$$x = 1$$

$g' = f$ changes from increasing to decreasing at $x = 1$.

Part C

2 points are earned for the correct values

<-1> each missing or extra value

$$x = -1, 1, 3$$



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the correct values

6-4

<-1> each missing or extra value

$$x = -1, 1, 3$$

Part D

1 point is earned for interval

1 point is earned for reason

h is decreasing on $[0, 2]$

$$h' = -f < 0 \text{ when } f > 0$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for interval

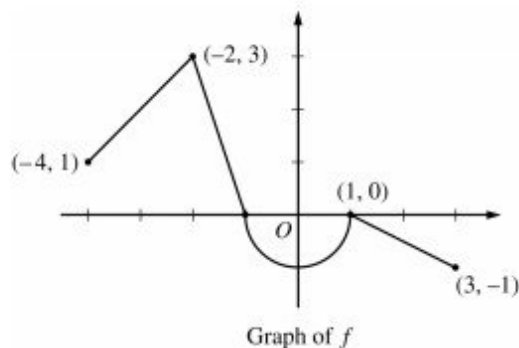
1 point is earned for reason

h is decreasing on $[0, 2]$

$$h' = -f < 0 \text{ when } f > 0$$

6-4

82.



Let f be the continuous function defined on $[-4, 3]$ whose graph, consisting of three line segments and a semicircle centered at the origin, is given above. Let g be the function given by $g(x) = \int_1^x f(t) dt$.

- Find the values of $g(2)$ and $g(-2)$.
- For each of $g'(-3)$ and $g''(-3)$, find the value or state that it does not exist.
- Find the x -coordinate of each point at which the graph of g has a horizontal tangent line. For each of these points, determine whether g has a relative minimum, relative maximum, or neither a minimum nor a maximum at the point. Justify your answers.
- For $-4 < x < 3$, find all values of x for which the graph of g has a point of inflection. Explain your reasoning.

Part A

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$\begin{aligned} g(-2) &= \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt \\ &= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right) \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(2)$

1 point is earned for $g(-2)$

6-4

$$g(2) = \int_1^2 f(t) dt = -\frac{1}{2}(1) \left(\frac{1}{2}\right) = -\frac{1}{4}$$

$$g(-2) = \int_1^{-2} f(t) dt = -\int_{-2}^1 f(t) dt$$

$$= -\left(\frac{3}{2} - \frac{\pi}{2}\right) = \left(\frac{\pi}{2} - \frac{3}{2}\right)$$

Part B1 point is earned for $g'(-3)$ 1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(-3)$ 1 point is earned for $g''(-3)$

$$g'(x) = f(x) \Rightarrow g'(-3) = f(-3) = 2$$

$$g''(x) = f'(x) \Rightarrow g''(-3) = f'(-3) = 1$$

Part C1 point is earned if considers $g'(x) = 0$ 1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$. $g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$. $g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

6-4



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if considers $g'(x) = 0$

1 point is earned for $x = -1$ and $x = 1$

1 point is earned for answers with justification

The graph of g has a horizontal tangent line where $g'(x) = f(x) = 0$. This occurs at $x = -1$ and $x = 1$.

$g'(x)$ changes sign from positive to negative at $x = -1$. Therefore, g has a relative maximum at $x = -1$.

$g'(x)$ does not change sign at $x = 1$. Therefore, g has neither a relative maximum nor a relative minimum at $x = 1$.

Part D

1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.



0	1	2
---	---	---

The student response earns all of the following points:

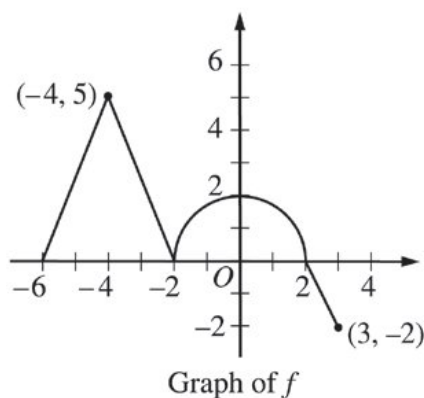
1 point is earned for the answer

1 point is earned for the explanation

The graph of g has a point of inflection at each of $x = -2$, $x = 0$, and $x = 1$ because $g''(x) = f'(x)$ changes sign at each of these values.

6-4

83.



The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above.

Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

- Find $g(-6)$ and $g(3)$.
- Find $g'(0)$.
- Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

Part A

The response can earn up to 2 points:

1 point is earned for $g(-6)$

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2} \pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$



0	1	2
---	---	---

The response earns all of the following points:

1 point is earned for $g(-6)$

6-4

$$g(-6) = \int_{-2}^{-6} f(t) dt = - \int_{-6}^{-2} f(t) dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$$

1 point is earned for $g(3)$

$$g(3) = \int_{-2}^3 f(t) dt = \frac{1}{2}\pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

Part B

The response can earn up to 1 point:

1 point is earned for the correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$



0	1
---	---

The response earns all of the following point:

1 point is earned for the correct evaluation of $g'(0)$

$$g'(0) = f(0) = 2$$

Part C

The response can earn up to 3 points:

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.



0	1	2	3
---	---	---	---

6-4

The response earns all of the following points:

1 point is earned for the horizontal tangent at $x = -2$ and $x = 2$

The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

2 points are earned for correct answers with justifications

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

Part D

The response can earn up to 3 points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.



0	1	2	3
---	---	---	---

The response earns all of the following points:

2 points are earned for the correct values of x

1 point is earned for the explanation

The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

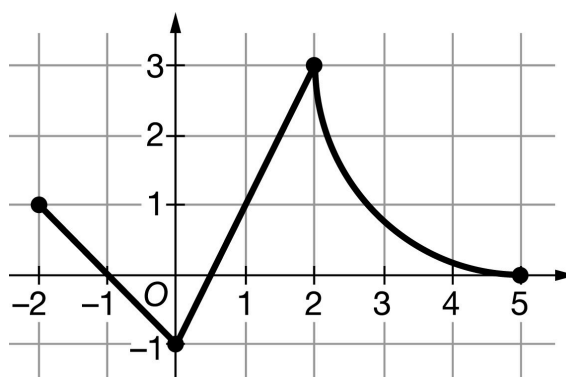
6-4

84. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $-6 \leq x \leq 5$. The figure above shows a portion of the graph of f , consisting of two line segments and a quarter of a circle centered at the point $(5, 3)$. It is known that the point $(3, 3 - \sqrt{5})$ is on the graph of f .

- If $\int_{-6}^5 f(x) \, dx = 7$, find the value of $\int_{-6}^{-2} f(x) \, dx$. Show the work that leads to your answer.
- Evaluate $\int_3^5 (2f'(x) + 4) \, dx$.
- The function g is given by $g(x) = \int_{-2}^x f(t) \, dt$. Find the absolute maximum value of g on the interval $-2 \leq x \leq 5$. Justify your answer.
- Find $\lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x}$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for showing evidence of using the property of adding integrals. This point can be earned symbolically or numerically with correct values of the integrals.

Simplification is not required to earn the second point or the third point

6-4



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $\int_{-6}^5 f(x) dx = \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx$ integral
- ☐ $\int_{-2}^5 f(x) dx$
- ☐ answer

Solution:

$$\begin{aligned}\int_{-6}^5 f(x) dx &= \int_{-6}^{-2} f(x) dx + \int_{-2}^5 f(x) dx \\ \Rightarrow 7 &= \int_{-6}^{-2} f(x) dx + 2 + \left(9 - \frac{9\pi}{4}\right) \\ \Rightarrow \int_{-6}^{-2} f(x) dx &= 7 - \left(11 - \frac{9\pi}{4}\right) = \frac{9\pi}{4} - 4\end{aligned}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an antiderivative of $f'(x)$ and use of the limits of integration. Responses with errors in the integrand, such as a missing coefficient, earn the first point; such errors are considered for the second point.

Simplification is not required to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Fundamental Theorem of Calculus
- ☐ answer

Solution:

6-4

$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= 2 \int_3^5 f'(x) \, dx + \int_3^5 4 \, dx \\
 &= 2(f(5) - f(3)) + 4(5 - 3) \\
 &= 2\left(0 - \left(3 - \sqrt{5}\right)\right) + 8 \\
 &= 2\left(-3 + \sqrt{5}\right) + 8 = 2 + 2\sqrt{5}
 \end{aligned}$$

OR

$$\begin{aligned}
 \int_3^5 (2f'(x) + 4) \, dx &= [2f(x) + 4x]_{x=3}^{x=5} \\
 &= (2f(5) + 20) - (2f(3) + 12) \\
 &= (2 \cdot 0 + 20) - \left(2\left(3 - \sqrt{5}\right) + 12\right) \\
 &= 2 + 2\sqrt{5}
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for connecting $g'(x) = f(x)$ analytically or verbally.

The second point is earned with identification of $x = -1$ regardless of what other candidates are considered.

The third point is earned for calculating $g(-1)$ and $g(5)$, declaring the absolute maximum to be $11 - \frac{9\pi}{4}$, and correctly reasoning for excluding $g(-2)$. Note that the ordered pair $(5, 11 - \frac{9\pi}{4})$ by itself is not a declaration of the absolute maximum.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ $g'(x) = f(x)$
- ☐ identifies $x = -1$ as a candidate
- ☐ answer with justification

Solution:

$$g'(x) = f(x) = 0 \Rightarrow x = -1, \quad x = \frac{1}{2}, \quad x = 5$$

6-4

x	$g(x)$
-2	0
-1	$\frac{1}{2}$
$\frac{1}{2}$	$-\frac{1}{4}$
5	$11 - \frac{9\pi}{4}$

On the interval $-2 \leq x \leq 5$, the absolute maximum value of g is $g(5) = 11 - \frac{9\pi}{4}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is earned for the correct answer. Simplification, including $\arctan 1 = \frac{\pi}{4}$, is not required.



0

1

The student response accurately includes the criteria below.



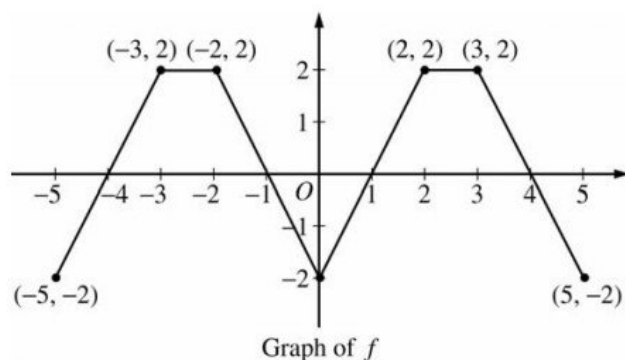
· answer

Solution:

$$\begin{aligned} \lim_{x \rightarrow 1} \frac{10^x - 3f'(x)}{f(x) - \arctan x} &= \frac{10^1 - 3f'(1)}{f(1) - \arctan 1} \\ &= \frac{10 - 3 \cdot 2}{1 - \arctan 1} = \frac{4}{1 - \frac{\pi}{4}} \end{aligned}$$

6-4

85.



The graph of the function f shown above consists of six line segments. Let g be the function given by

$$g(x) = \int_0^x f(t) dt.$$

(a) Find $g(4)$, $g'(4)$, and $g''(4)$.

(b) Does g have a relative minimum, a relative maximum, or neither at $x = 1$? Justify your answer.

(c) Suppose that f is defined for all real numbers x and is periodic with a period of length 5. The graph above shows two periods of f . Given that $g(5) = 2$, find $g(10)$ and write an equation for the line tangent to the graph of g at $x = 108$.

Part A

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g(4)$

1 point is earned for: $g'(4)$

1 point is earned for: $g''(4)$

6-4

$$g(4) = \int_0^4 f(t) dt = 3$$

$$g'(4) = f(4) = 0$$

$$g''(4) = f'(4) = -2$$

Part B

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

g has a relative minimum at $x = 1$

because $g' = f$ changes from negative to positive at $x = 1$.

Part C

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

6-4

$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for: $g(10)$

1 point is earned for: $g(108)$

1 point is earned for: $g'(108)$

1 point is earned for the equation of tangent line

$g(0) = 0$ and the function values of g increase by 2 for every increase of 5 in x .

$$g(10) = 2g(5) = 4$$

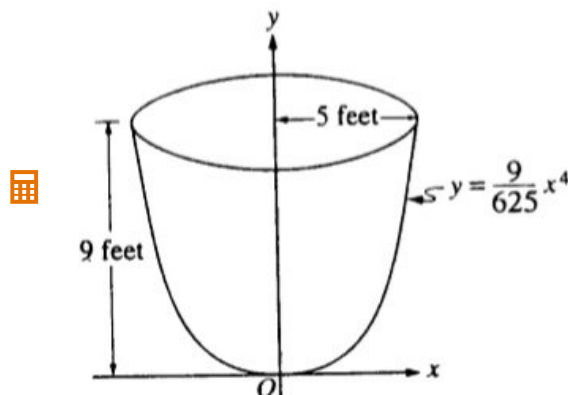
$$\begin{aligned}
 g(108) &= \int_0^{105} f(t)dt + \int_{105}^{108} f(t)dt \\
 &= 21g(5) + g(3) = 44
 \end{aligned}$$

$$g'(108) = f(108) = f(3) = 2$$

An equation for the line tangent to the graph of g at $x = 108$ is $y - 44 = 2(x - 108)$.

6-4

86.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil flows into the tank at the constant rate of 8 cubic feet per minute.

(a) Find the volume of the tank. Indicate units of measure.

(b) To the nearest minute, how long would it take to fill the tank if the tank was empty initially?

(c) Let h be the depth, in feet, of oil in the tank. How fast is the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

Part A

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \pi \int_0^9 (\) dy, 2\pi \int_0^5 (\) dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi \text{ ft}^3$$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi \int_0^5 x \left(9 - \frac{9}{625}x^4 \right) dx = 150\pi \text{ ft}^3$$

[465, 475] [148π , 152π]

6-4



0	1	2	3
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The student response earns all of the following points:

2 points are earned for the correct integral where one point is earned for the correct limits and

$$\frac{\pi}{2\pi} \pi \int_0^9 () dy, 2\pi \int_0^5 () dx$$

and one point is earned for the correct integrand

$$\text{Volume} = V = \pi = \int_0^9 \frac{25}{3} \sqrt{y} dy = 150\pi \text{ ft}^3$$

(or 471.238 ft^3 or 471.239 ft^3)

1 point is earned for the correct answer with units

Note: not eligible for answer point unless first two points are earned

$$V = 2\pi = \int_0^5 x \left(9 - \frac{9}{625} x^4 \right) dx = 150\pi \text{ ft}^3$$

$[465, 475]$ $[148\pi, 152\pi]$

Part B

1 point is earned for the correct integer answer rounded up or down

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct integer answer rounded up or down

6-4

$$\text{time} = \frac{\text{their Volume}}{\text{rate}} = \frac{150\pi}{8}$$

Therefore, 59 minutes

it is not necessary to indicate unit of measure

Part C

1 point is earned for the correct volume as definite integral using h

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)

if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for the correct volume as definite integral using h

6-4

Finding $\frac{dV}{dt}$

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} dy = \frac{50\pi}{9} h^{3/2}$$

1 point is earned for the correct $\frac{dV}{dh}$ 1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ by chain rule1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\frac{dV}{dt} = \frac{25}{3} \pi \sqrt{h} \frac{dh}{dt}$$

$$\frac{dV}{dt} = 8$$

$$\text{when } h = 4, 8 = \frac{25}{3} \pi (2) \frac{dh}{dt}$$

1 point is earned for the correct answer with units

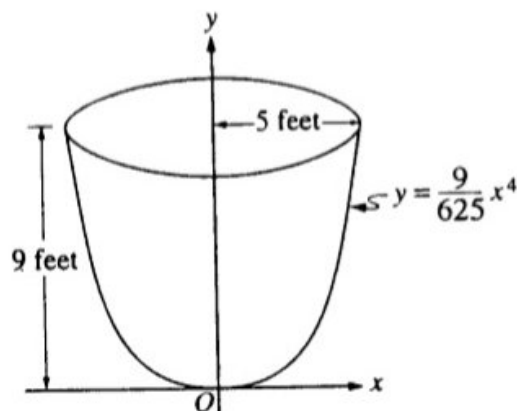
Note: if V is linear, max 2/5 (0 - 0 - 1 - 1 - 0)if V is constant, max 1/5 (0 - 0 - 0 - 1 - 0)

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

(or 0.152 ft/min or 0.153 ft/min)

6-4

87.



An oil storage tank has the shape as shown above, obtained by revolving the curve $y = \frac{9}{625}x^4$ from $x = 0$ to $x = 5$ about the y -axis, where x and y are measured in feet.

Oil weighing 50 pounds per cubic foot flowed into an initially empty tank at a constant rate of 8 cubic feet per minute. When the depth of oil reached 6 feet, the flow stopped.

(a) Let h be the depth, in feet, of oil in the tank. How fast was the depth of oil in the tank increasing when $h = 4$? Indicate units of measure.

(b) Find, to the nearest foot-pound, the amount of work required to empty the tank by pumping all of the oil back to the top of the tank.

Part A

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} \, dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

6-4

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)



0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for the correct volume as integral with h

$$V = \pi \int_0^h \frac{25}{3} \sqrt{y} \, dy$$

Finding $\frac{dV}{dt}$:

1 point is earned for the correct $\frac{dV}{dh}$

1 point is earned for the correct $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$

$$\frac{dV}{dt} = \frac{25\pi}{3} \sqrt{h} \frac{dh}{dt}$$

1 point is earned for the correct answer $\frac{dV}{dt} = 8$

$$\text{at } h = 4, 8 = \frac{25\pi}{3} \sqrt{4} \frac{dh}{dt}$$

1 point is earned for the correct answer with units

$$\frac{dh}{dt} = \frac{12}{25\pi} \text{ ft/min}$$

Note: If V is linear, max 2/5 (0 0 1 1 0)

If V is constant, max 1/5 (0 0 0 1 0)

Part B

1 point is earned for the correct limits

$$W = 50 \int_0^6 (9 - y) \left(\frac{25\pi}{3} \sqrt{y} \right) dy$$

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

6-4

$$W = 50 \left(\frac{25\pi}{3} \right) \int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}} \right) dy$$

$$W = 50 \left(\frac{25\pi}{3} \right) \left(\frac{2}{3} \cdot 9y^{\frac{3}{2}} - \frac{2}{5} y^{\frac{5}{2}} \right) \Big|_0^6$$

1 point is earned for the correct answer

$W = 69,257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits

$$W = 50 \int_0^6 (9 - y) \left(\frac{25\pi}{3} \sqrt{y} \right) dy$$

2 points are earned for integrand where 1 point is earned for the force and 1 point is earned for the distance

$$W = 50 \left(\frac{25\pi}{3} \right) \int_0^6 \left(9y^{\frac{1}{2}} - y^{\frac{3}{2}} \right) dy$$

$$W = 50 \left(\frac{25\pi}{3} \right) \left(\frac{2}{3} \cdot 9y^{\frac{3}{2}} - \frac{2}{5} y^{\frac{5}{2}} \right) \Big|_0^6$$

1 point is earned for the correct answer

$W = 69,257.691$ ft-lbs to the nearest foot-pound 69,258 ft-lbs

(0/1 if no integrand points)

6-4

88.  A particle moves along the x -axis so that its velocity at any time $t \geq 0$ is given by $v(t) = 3t^2 - 2t - 1$. The position $x(t)$ is 5 for $t = 2$.
- (a) Write a polynomial expression for the position of the particle at any time $t \geq 0$.
- (b) For what values of t , $0 \leq t \leq 3$, is the particle's instantaneous velocity the same as its average velocity on the closed interval $[0, 3]$?
- (c) Find the total distance traveled by the particle from time $t = 0$ until time $t = 3$.

Part A

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; \quad C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct answer $x(t) = t^3 - t^2 - t + C$

<-1> for error(s) in $t^3 - t^2 - t$

<-1> for no constant of integration

$$\begin{aligned} x(t) &= \int v(t) dt = \int (3t^2 - 2t - 1) dt \\ &= t^3 - t^2 - t + C \end{aligned}$$

6-4

1 point is earned for the correct evaluation of the constant of integration

$$x(2) = 8 - 4 - 2 + C = 5; C = 3$$

$$x(t) = t^3 - t^2 - t + 3$$

Part B

1 point is earned for the correct average velocity = 5

$$\begin{aligned}\text{avg. vel.} &= \frac{x(3) - x(0)}{3 - 0} \\ &= \frac{18 - 3}{3} = 5\end{aligned}$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

0/1 if not solving $3t^2 - 2t - 1 = \text{an avg. velocity in } [0, 3]$

$$t = \frac{1 + \sqrt{19}}{3} \text{ or } 1.786$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct average velocity = 5

$$\begin{aligned}\text{avg. vel.} &= \frac{x(3) - x(0)}{3 - 0} \\ &= \frac{18 - 3}{3} = 5\end{aligned}$$

1 point is earned for correctly setting $v(t)$ equal to student's average velocity OR from student's $x(t)$

$$3t^2 - 2t - 1 = 5$$

1 point is earned for the correct answer

6-4

0/1 if not solving $3t^2 - 2t - 1 = 0$ as an avg. velocity in $[0, 3]$

$$t = \frac{1+\sqrt{19}}{3} \text{ or } 1.786$$

Part C

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\begin{aligned} \text{distance} &= \int_0^3 |v(t)| dt \\ &= \int_0^3 |3t^2 - 2t - 1| dt = 17 \end{aligned}$$

or

$$v(t) = 3t^2 - 2t - 1 = 0$$

$$t = -\frac{1}{3}, t = 1$$

$$x(0) = 3$$

$$x(1) = 1 - 1 - 1 + 3 = 2$$

$$x(3) = 27 - 9 - 3 + 3 = 18$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

$$\text{distance} = (3 - 2) + (18 - 2) = 17$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits of 0 and 3 on an integral of $v(t)$ or $|v(t)|$ or substitutes $t = 0$ and $t = 3$ in $x(t)$

$$\begin{aligned} \text{distance} &= \int_0^3 |v(t)| dt \\ &= \int_0^3 |3t^2 - 2t - 1| dt = 17 \end{aligned}$$

6-4

or

$$v(t) = 3t^2 - 2t - 1 = 0$$

$$t = -\frac{1}{3}, t = 1$$

$$x(0) = 3$$

$$x(1) = 1 - 1 - 1 + 3 = 2$$

$$x(3) = 27 - 9 - 3 + 3 = 18$$

1 point is earned for the correct handling of the change in direction at student's turning point

1 point is earned for the correct answer

$$\text{distance} = (3 - 2) + (18 - 2) = 17$$

89. If $h(x) = \int_{-\frac{\pi}{4}}^{\sin^2 x} \csc(t^4 + 1) dt$ for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$, then $h'(x) =$

(A) $\csc(x^4 + 1)$

(B) $\csc(\sin^8 x + 1)$

(C) $(\sin^2 x)(\csc(x^4 + 1))$

(D) $(2 \sin x \cos x)(\csc(\sin^8 x + 1))$



Answer D

Correct. Let $f(x) = \int_{-\frac{\pi}{4}}^x \csc(t^4 + 1) dt$ and $g(x) = \sin^2 x$. Then

$h(x) = \int_{-\frac{\pi}{4}}^{\sin^2 x} \csc(t^4 + 1) dt = f(g(x))$. By the chain rule, $h'(x) = f'(g(x))g'(x)$, where $f'(x) = \csc(x^4 + 1)$ and $g'(x) = 2 \sin x \cos x$.

6-4

90. For time $t \geq 0$, the position of a particle moving along a straight line is given by

$$s(t) = \int_0^t e^{\sin x} \cos x \, dx. \text{ What is the velocity of the particle at time } t = 3?$$

(A) $e^{\sin 3} - 1$

(B) $e^{\sin 3} \cos 3$ ✓

(C) $e^{\sin 3} \cos 3 - 1$

(D) $e^{\sin 3}(\cos 3)^2 - e^{\sin 3} \sin 3$

Answer B

Correct. The velocity of the particle is the derivative of the particle's position.

$$v(t) = s'(t) = \frac{d}{dt} \int_0^t e^{\sin x} \cos x \, dx = e^{\sin t} \cos t$$

$$v(3) = e^{\sin 3} \cos 3$$

91. If $f(x) = \int_0^{3x} e^{t^2} \, dt$, then $f'(x) =$

(A) e^{x^2}

(B) e^{9x^2}

(C) $3e^{9x^2}$ ✓

(D) $18xe^{9x^2}$

Answer C

Correct. Let $g(x) = \int_0^x e^{t^2} \, dt$. Then $g'(x) = e^{x^2}$. Because $f(x) = g(3x)$, the chain rule gives

$$f'(x) = g'(3x) \cdot 3 = e^{(3x)^2} \cdot 3 = 3e^{9x^2}.$$

6-4

92. $\frac{d}{dx} \left(\int_1^x t^4 + t^2 + 1 \, dt \right) =$

(A) $x^4 + x^2 + 1$ ✓

(B) $4x^3 + 2x$

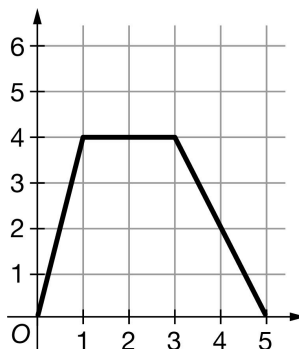
(C) $x^4 + x^2 - 2$

(D) $4x^3 + 2x - 6$

Answer A

Correct. If f is a continuous function on an interval containing a , then $\frac{d}{dx} \left(\int_a^x f(t) \, dt \right) = f(x)$. Here $f(t) = t^4 + t^2 + 1$ and $a = 1$.

93.



Graph of f

The graph of the function f shown above consists of three line segments. If h is the function defined by

$h(x) = \int_0^x f(t) \, dt$ for $0 \leq x \leq 5$, then $h'(3)$ is

(A) 0

(B) 4 ✓

(C) 10

(D) undefined

Answer B

Correct. If f is a continuous function on an interval containing a , then $\frac{d}{dx} \left(\int_a^x f(t) \, dt \right) = f(x)$. Because

$h(x) = \int_0^x f(t) \, dt$, it follows that $h'(x) = f(x)$ and thus $h'(3) = f(3) = 4$.

6-4

94. $\frac{d}{dx} \left(\int_2^x \sin(t^4) dt \right) =$

(A) $\sin(x^4)$



(B) $4x^3 \cos(x^4)$

(C) $\sin(x^4) - \sin(16)$

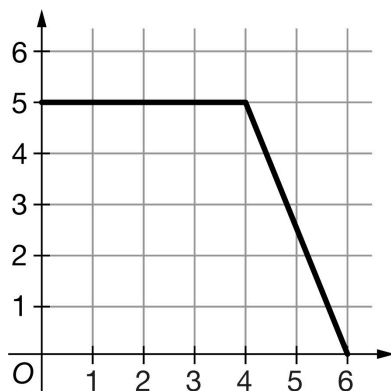
(D) $4x^3 \cos(x^4) - 32 \cos(16)$

Answer A

Correct. If f is a continuous function on an interval containing a , then $\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$. Here $f(t) = \sin(t^4)$ and $a = 2$.

6-4

95.

Graph of f

The graph of the function f , shown above, consists of two line segments. If h is the function defined by

$$h(x) = \int_0^x f(t) \, dt \text{ for } 0 \leq x \leq 6, \text{ then } h'(4) \text{ is}$$

- (A) 0
- (B) 5
- (C) 20
- (D) undefined

**Answer B**

Correct. If f is a continuous function on an interval containing a , then $\frac{d}{dx} \left(\int_a^x f(t) \, dt \right) = f(x)$. Because

$h(x) = \int_0^x f(t) \, dt$, it follows that $h'(x) = f(x)$ and thus $h'(4) = f(4) = 5$.

6-4

96. If $h(x) = \int_{-1}^{x^3} \sqrt{2+t^2} \, dt$ for $x \geq 0$, then $h'(x) =$

(A) $\sqrt{2+x^2}$

(B) $\sqrt{2+x^6}$

(C) $3x^2\sqrt{2+x^2}$

(D) $3x^2\sqrt{2+x^6}$

**Answer D**

Correct. Let $f(x) = \int_{-1}^x \sqrt{2+t^2} \, dt$ and $g(x) = x^3$. Then $h(x) = \int_{-1}^{x^3} \sqrt{2+t^2} \, dt = f(g(x))$. By the chain rule, $h'(x) = f'(g(x))g'(x)$, where $f'(x) = \sqrt{2+x^2}$ and $g'(x) = 3x^2$.

97. Let f be the function defined by $f(x) = 4 + \int_1^x \frac{1}{1+t} \, dt$. Which of the following is an equation for the line tangent to the graph of f at $x = 1$?

(A) $y = 4$

(B) $y - 4 = \frac{1}{2}(x - 1)$

(C) $y - 4 = \frac{1}{4}(x - 1)$

(D) $y - 4 = \frac{9}{2}(x - 1)$

**Answer B**

Correct. An equation of the line tangent to the graph of f at $x = a$ is $y - f(a) = f'(a)(x - a)$. Here $a = 1$.

$$f(1) = 4 + \int_1^1 \frac{1}{1+t} \, dt = 4 + 0 = 4$$

$$f'(x) = \frac{d}{dx} \left(4 + \int_1^x \frac{1}{1+t} \, dt \right) = \frac{1}{1+x} \Rightarrow f'(1) = \frac{1}{2}$$

An equation of the tangent line is therefore $y - 4 = \frac{1}{2}(x - 1)$.

6-4

98.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	6	4	2	5
2	9	2	3	1
3	10	-4	4	2
4	-1	3	6	7

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

(b) Explain why there must be a value c for $1 < c < 3$ such that $h'(c) = -5$.

(c) Let w be the function given by $w(x) = \int_1^{g(x)} f(t) dt$. Find the value of $w'(3)$.

(d) If g^{-1} is the inverse function of g , write an equation for the line tangent to the graph of $y = g^{-1}(x)$ at $x = 2$.

Part A

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $h(1)$ and $h(3)$

1 point is earned for the conclusion, using IVT

$$h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

6-4

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$.

Part B

1 point is earned for: $\frac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3) - h(1)}{3 - 1} = \frac{-7 - 3}{3 - 1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{h(3) - h(1)}{3 - 1}$

1 point is earned for the conclusion, using MVT

$$\frac{h(3) - h(1)}{3 - 1} = \frac{-7 - 3}{3 - 1} = -5$$

Since h is continuous and differentiable, by the Mean Value Theorem, there exists a value c , $1 < c < 3$, such that $h'(c) = -5$.

Part C

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3) = f(g(3)) \cdot g'(3) = f(4) \cdot 2 = -2$$

6-4



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for applying chain rule

1 point is earned for correct answer

$$w'(3) = f(g(3)) \cdot g'(3) = f(4) \cdot 2 = -2$$

Part D

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

$$g(1) = 2, \text{ so } g^{-1}(2) = 1.$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g^{-1}(2)$

1 point is earned for $(g^{-1})'(2)$

1 point is earned for the tangent line equation

6-4

$$g(1) = 2, \text{ so } g^{-1}(2) = 1.$$

$$(g^{-1})'(2) = \frac{1}{g'(g^{-1}(2))} = \frac{1}{g'(1)} = \frac{1}{5}$$

An equation of the tangent line is $y - 1 = \frac{1}{5}(x - 2)$.

6-4

99. Water is pumped into an underground tank at a constant rate of 8 gallons per minute. Water leaks out of the tank at the rate of $\sqrt{t+1}$ gallons per minute, for $0 \leq t \leq 120$ minutes. At time $t = 0$, the tank contains 30 gallons of water.
- How many gallons of water leak out of the tank from time $t = 0$ to $t = 3$ minutes?
 - How many gallons of water are in the tank at time $t = 3$ minutes?
 - Write an expression for $A(t)$, the total number of gallons of water in the tank at time t .
 - At what time t , for $0 \leq t \leq 120$, is the amount of water in the tank a maximum? Justify your answer.

Part AMethod 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; L(3) = \frac{14}{3}$$

6-4



0	1	2	3	4	5	6
---	---	---	---	---	---	---

The student response earns all of the following points:

Method 1:

2 points are earned for the definite integral

1: limits

1: integrand

1 point is earned for the answer

- or -

Method 2:

1 point is earned for antiderivative with C

1 point is earned for solves for C using $L(0) = 0$

1 point is earned for the answer

Method 1: $\int_0^3 \sqrt{t+1} \, dt = \frac{2}{3}(t+1)^{3/2} \Big|_0^3 = \frac{14}{3}$

- or -

Method 2: $L(t)$ = gallons leaked in first t minutes

$$\frac{dL}{dt} = \sqrt{t+1}; L(t) = \frac{2}{3}(t+1)^{3/2} + C$$

$$L(0) = 0; C = -\frac{2}{3}$$

$$L(t) = \frac{2}{3}(t+1)^{3/2} - \frac{2}{3}; L(3) = \frac{14}{3}$$

Part B

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

6-4



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$30 + 8 \cdot 3 - \frac{14}{3} = \frac{148}{3}$$

Part C

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

6-4

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + C$$

$$30 = 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for: $30 + 8t$

1 point is earned for: $-\int_0^t \sqrt{x+1} \, dx$

- or -

Method 2:

1 point earned for antiderivative with C

1 point is earned for the answer

Method 1:

$$\begin{aligned} A(t) &= 30 + \int_0^t (8 - \sqrt{x+1}) \, dx \\ &= 30 + 8t - \int_0^t \sqrt{x+1} \, dx \end{aligned}$$

- or -

Method 2:

6-4

$$\frac{dA}{dt} = 8 - \sqrt{t+1}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + C$$

$$30 = 8(0) - \frac{2}{3}(0+1)^{3/2} + C; \quad C = \frac{92}{3}$$

$$A(t) = 8t - \frac{2}{3}(t+1)^{3/2} + \frac{92}{3}$$

Part D

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for set $A'(t) = 0$

1 point is earned for solves for t

1 point is earned for the justification

$$A'(t) = 8 - \sqrt{t+1} = 0 \text{ when } t = 63$$

$A'(t)$ is positive for $0 < t < 63$ and negative for $63 < t < 120$. Therefore there is a maximum at $t = 63$.